

From one cell to a population

Burst-aware renewal dynamics and the bursting–budding comparison

Adam Aldridge

University of Leeds

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Chapter 1

From one cell to a population: burst-aware renewal dynamics and the bursting–budding comparison

1.1 Introduction

A fitted release rate is not a property of a pathogen. In the standard within-host model an infected cell sheds virions at a constant rate p and is removed at a constant rate d_I , and both numbers are read off from data as though they characterised the organism. For a budding virus that is a defensible caricature: release and death really are separate, independent events, and two constants do capture the cell. For a lytic pathogen it is wrong in mechanism. Such a pathogen releases nothing while it grows inside the cell, and then releases everything at once, in the event that kills the cell. Release and death are not independent; they are the same event. *Y. pestis* in the macrophage, *F. tularensis* in the macrophage, the lytic phage in its bacterium — all are of that kind, and none is described by a constant drip. Nor, once the mechanism is put back in, does p survive as a constant of the organism: it becomes a property of the phase the infection is in.

The chapter on the birth–death–catastrophe process set up the intracellular process for such a pathogen, a birth–death–catastrophe process with rates λ, μ, δ ; the chapter on its distribution theory completed its distribution theory. Both stop at the cell boundary. This chapter carries the single-cell theory across it, and asks what classical BMVR becomes once the cell inside it has been given a mechanism.

1.1.1 Four questions, and their answers in advance

1. What replaces the release term $p\mathcal{I}$ when release is terminal and batched?
2. In what precise sense is the classical model still right, and how wrong is it elsewhere?
3. Does bursting help or hurt a pathogen, and at what cost?
4. How much of any of this depends on the birth–death–catastrophe process in particular?

The answers, in brief. What replaces $p\mathcal{I}$ is a renewal system: infection events start cohorts, cohorts age, and two kernels weight them — the probability $S(\alpha)$ that a cell of age α is still productive, and the expected release flux $g(\alpha)$ at that age. Both are already in closed form from the single-cell theory, so the population model is forced by the intracellular process rather than fitted alongside it.

The classical model is recovered exactly, but one growth rate at a time. For each r there is a unique constant pair $(p_{\text{eff}}(r), d_{\mathcal{I},\text{eff}}(r))$ that reproduces the renewal system's exponential solution exactly; what no single pair reproduces is a transient, which samples a continuum of growth rates at once. Across that map the release rate moves by two orders of magnitude. At $(\lambda, \mu, \delta) = (1, 0, 0.1)$ the same cell presents $p_{\text{eff}} = 4.59$ in a stationary infection and $p_{\text{eff}} \rightarrow \delta = 0.1$ in a fast-growing one. No fit of $(p, d_{\mathcal{I}})$ recovers the three intracellular rates: the map from (λ, μ, δ) to a constant pair has a kernel, and a third observable is required before a population fit can test the intracellular model.

Bursting does both. At fixed total output per cell it leaves R_0 exactly where it was, lowers the establishment extinction probability when $1/\delta < 1/\lambda + 1/\mu$, and slows deterministic invasion in every regime examined. Threshold, establishment risk and invasion speed are three different questions, and bursting answers them differently.

Almost none of it depends on the birth–death–catastrophe process. The population equations use that process only through the pair (S, g) it supplies, so any single-cell mechanism supplying such a pair induces the same equations, the same effective-parameter map and the same reading of R_0 and r : bursting and budding are two choices of one object, not two theories. What does depend on the particular process is everything resting on its closed forms — the flooding criterion above, the geometric burst law, the endpoints of the map. [Section 1.6](#) draws that line.

1.1.2 Standing assumptions

These hold throughout unless stated otherwise.

- (A1) Target cells T are held constant: no depletion, no saturation, no immune response.
- (A2) Virions are not consumed on infection — the *continuum* convention. This is why the deterministic threshold is $R_0 = \gamma T V_{\infty} / c$ while the discrete branching mean of [section 1.5](#) is $m = q V_{\infty}$ with $q = \gamma T / (\gamma T + c)$; the two agree when $\gamma T \ll c$.
- (A3) The $\mathcal{I}_0$ cells present at $t = 0$ all have age exactly zero. A non-trivial initial age distribution adds a term to [\(1.10\)](#).
- (A4) Infected cells are independent, and the population layer assumes single infection: MOI- k kernels are available ([section 1.A](#)) but are not used.
- (A5) There is no spatial structure.

1.1.3 Notation

Throughout, λ , μ and δ are the intracellular birth, death and catastrophe rates per replicative unit. The roots a, b , the derived constants A, B, θ, κ , and the single-cell functions $I, D, I_{\text{fix}}, J, K, V$ are those of the chapter on the birth–death–catastrophe process, quoted in [section 1.A](#). One naming decision is made now and kept: the burst-time density, written $\varphi(t)$ in the chapter on its distribution theory, is written δJ here, because φ is reserved for the partial-release fraction of [section 1.6.3](#). A second follows it: the survival kernel S of [section 1.3](#) is the non-fixation probability I_{fix} , written S where it acts as a kernel and I_{fix} where the closed form is meant. Lifetime yield V_{∞} and mean productive lifetime $\mathbb{E}[T_{\text{prod}}]$ are used from the first calculation onward; their closed forms sit with the rest in [section 1.A](#). [Table 1.1](#) fixes the population symbols.

Symbol	Meaning	Notes
α	cell age	argument of every kernel
T, γ, c	target cells, infection rate, virion clearance	T held fixed
$\mathcal{I}, \mathcal{V}$	infected cells, free particles	population counts
$i(t)$	incidence, $\gamma T \mathcal{V}$	
$d_{\mathcal{I}}$	infected-cell removal rate	phenomenological
$p, p_{\text{eff}}(r)$	constant, effective release rate	
$S(\alpha), g(\alpha)$	survival kernel $I_{\text{fix}}(\alpha)$, release kernel $\delta K(\alpha)$	
$\mathcal{A}(\alpha)$	generation kernel	(1.23)
r	epidemic growth rate	
q, m	infection success $\gamma T / (\gamma T + c)$, branching mean qV_{∞}	
L	flooding parameter $a(1 - b)$	
ν, ε, Q	production, export rates, total stock	section 1.6.1
$\bar{x}$	mean load per productive cell	$Q/\mathcal{I}$
ω, d_E, E	eclipse conversion, eclipse death, eclipse class	
φ	partial-release fraction	section 1.6.3
ψ, ζ, η	release intensity, its decay and jump sizes	section 1.E

Table 1.1: Population notation of this chapter. Single-cell symbols are those of [section 1.A](#).

1.2 The comparator and the obstruction

The single-cell theory is complete; what it implies for a population of infected cells is not. The obstruction shows itself most clearly against the standard model this chapter is meant to improve upon, which is where we begin.

1.2.1 The budding cell

In the simplified, target-cell-constant Basic Model of Viral Replication [1, 2, 3], infected cells $\mathcal{I}$ and free particles $\mathcal{V}$ obey

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (1.1)$$

with target cells T fixed, infection rate γ , virion clearance c , and the two constants that concern us: p , the per-cell release rate, and $d_{\mathcal{I}}$, the removal rate. Both are phenomenological, parameters to be fitted rather than quantities derived from anything inside the cell. The basic reproduction number is

$$R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (1.2)$$

Every cell, regardless of its age or its contents, releases at rate p and dies at rate $d_{\mathcal{I}}$.

That is not an arbitrary caricature. It is one kind of cell described exactly, and it is worth setting out in full, because it is the comparator the whole chapter is built against. A budding infected cell experiences two *independent* event types: release of one particle, as a Poisson process of rate p , and death, as a Poisson process of rate $d_{\mathcal{I}}$; the two run side by side until the cell dies. Its survival and cumulative release are

$$I_{\text{bud}}(\alpha) = e^{-d_{\mathcal{I}}\alpha}, \quad V_{\text{bud}}(\alpha) = \frac{p}{d_{\mathcal{I}}} \left(1 - e^{-d_{\mathcal{I}}\alpha}\right), \quad \frac{dV_{\text{bud}}}{dI_{\text{bud}}} = -\frac{p}{d_{\mathcal{I}}} : \quad (1.3)$$

the mean lifetime is $1/d_{\mathcal{I}}$, the lifetime release $p/d_{\mathcal{I}}$, and the total released is geometric on $\{0, 1, 2, \dots\}$ with $\mathbb{P}\{\mathcal{K}_{\text{bud}} = k\} = \frac{d_{\mathcal{I}}}{d_{\mathcal{I}} + p} \left(\frac{p}{d_{\mathcal{I}} + p}\right)^k$ — the successes before the terminating death in

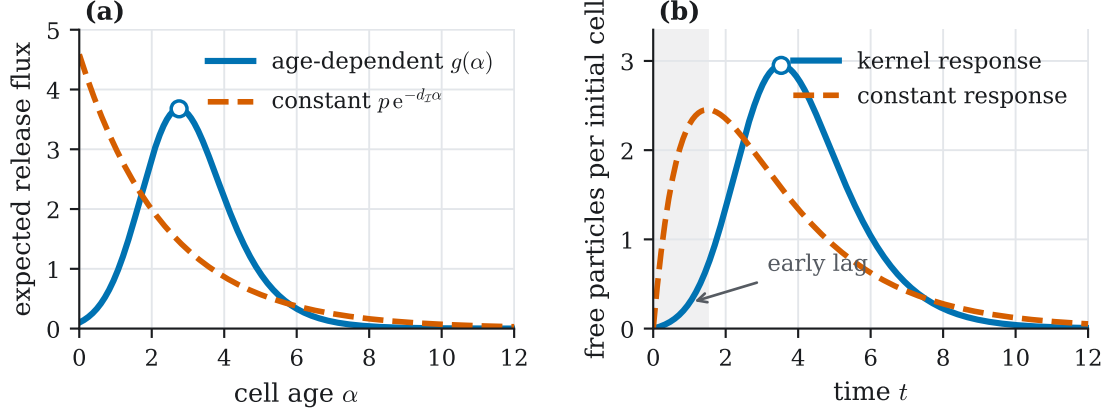


Figure 1.1: A synchronous cohort releases on a schedule, not at a rate. Both schedules have lifetime yield $V_\infty = 11$ and mean productive lifetime $\mathbb{E}[T_{\text{prod}}] = 2.398$, with $d_I = 1/\mathbb{E}[T_{\text{prod}}]$, $p = V_\infty d_I$, $(\lambda, \mu, \delta) = (1, 0, 0.1)$ and $c = 1$: as fairly matched as two constants allow. (a) Matched yield, different release timing: the release kernel $g(\alpha) = \delta K(\alpha)$ still lags the budding cell's $pe^{-d_I\alpha}$, because a bursting cell must first accumulate load. (b) Linear clearance preserves that lag, carrying it into the free-particle response.

a sequence of events each of which is a release with probability $p/(d_I + p)$. All of it sits in the pair (p, d_I) . That is why (1.1) is a faithful one-cell limit for a budding pathogen, and it is this cell that section 1.5 matches against a bursting one.

The structural contrast with bursting is the engine of everything that follows. In budding, release and death are independent: a cell may release many particles and then die with its contents lost, or release nothing at all. In bursting they are the *same* event: whatever has accumulated is released at once, and the cell's productive life ends in that instant. The detailed single-cell comparison is in the chapter on its distribution theory; here the point is that (1.1) encodes the budding picture exactly, and the question is what replaces it when release comes in terminal batches.

1.2.2 The synchronous cohort

Take a clean special case: N_0 cells all infected at the same instant, each running an independent copy of the single-cell process, with no reabsorption or extracellular replication. The number still productively infected at time t is $N_0 I_{\text{fix}}(t)$, exactly in expectation, and the expected cumulative extracellular count is

$$\mathcal{V}(t) = N_0 V(t), \quad (1.4)$$

since $V(t)$ is the expected release from one cell. The release *rate* is then $N_0 V'(t) = N_0 \delta K(t)$: it depends on the age of the cohort, and it is not constant. The classical model's $p\mathcal{I}$ can mimic such a cohort only by accident, and only briefly (figure 1.1).

1.2.3 The proposal that cannot work

The natural thought is to replace the constant p in (1.1) by a constant built from the intracellular theory: the quasi-stationary mean,

$$p \stackrel{?}{=} \langle X \rangle_{\text{QS}} = \frac{a}{a-1}. \quad (1.5)$$

The intuition is that a long-lived cell hovers at the quasi-stationary level, so its bursting should look like release at rate $\langle X \rangle_{\text{QS}}$. The obstruction is dimensional: p has units of particles per unit

time and $\langle X \rangle_{\text{QS}}$ is a pure count, so (1.5) introduces a spurious factor of time. Worse, when $\mu = 0$ one has $\langle X \rangle_{\text{QS}} = V_\infty$ exactly, so the proposal substitutes a cell's *total lifetime output* for its *output rate*; figure 1.19 shows what that costs.

Two repairs suggest themselves, and both fail for related reasons. First, average the burst size over the burst-time density δJ of the chapter on its distribution theory:

$$\int_0^\infty \delta J(t) \mathbb{E}[\mathcal{K} \mid \tau = t] dt = \delta \int_0^\infty K(t) dt = V_\infty, \quad (1.6)$$

using (1.60) and $V' = \delta K$ — again a count, not a rate. Second, weight the load-dependent flux by the burst-time density, $\delta \int \delta J(t) \mathbb{E}[X_t^2 \mid \cdot] dt$: the dimensions are right, but the measure is wrong. The burst-time density describes *when a single cell bursts*; a population release rate needs an average over the *ages of the cells currently infected*, and in an asynchronous infection those are not the same. No reweighting of a single-cell count by a single-cell density can fix the substitution, because the difficulty is not in the single-cell theory at all. It is in the population bookkeeping.

What survives the failed proposal is two facts, not a substitute constant. The expected release flux of a cell of age α is $\delta K(\alpha)$, and at any host time the infected population is a mixture of ages, one cohort per past moment of infection. The population model must therefore track cell age. That is the renewal construction of section 1.3.

1.2.4 Three regimes, used throughout

Three intracellular parameter triples recur throughout, chosen to straddle the flooding boundary $L = 1$ of section 1.5: one on each side, one exactly on it. They are named here and used by name thereafter, and table 1.2 collects what follows from them by (1.49)–(1.57).

(λ, μ, δ)	a	b	L	V_∞	$\mathbb{E}[T_{\text{prod}}]$	$d_{\mathcal{I},\text{eff}}(0)$	$p_{\text{eff}}(0)$
(1, 0, 0.1)	1.100	0	1.10	11.00	2.398	0.417	4.59
(1, 0.5, 1/3)	1.500	1/3	1.00	2.00	1.099	0.910	1.82
(1, 0.9, 0.1)	1.316	0.684	0.42	1.32	1.426	0.701	0.92

Table 1.2: The three regimes used throughout, and what follows from them: the first has $L > 1$, the second sits on the boundary $L = 1$, the third has $L < 1$. The last two columns are the $r = 0$ endpoints of the effective-parameter map of section 1.4. The verification suite of section 1.C uses these three with (1, 0.2, 0.05) and (1, 0, 0.5); a wider spread suits testing than exposition.

1.3 Burst-aware renewal dynamics

1.3.1 The two kernels

Age-structured population dynamics needs exactly two single-cell ingredients, and both are already in closed form. The *survival kernel* is the probability that a cell infected α time units ago is still productively infected:

$$S(\alpha) := I_{\text{fix}}(\alpha) = \frac{(a - b)^2 e^{\theta\alpha}}{(B + Ae^{\theta\alpha})(ae^{\theta\alpha} - b)}. \quad (1.7)$$

The *release kernel* is the expected release flux at cell age α :

$$g(\alpha) := \delta K(\alpha) = V'(\alpha). \quad (1.8)$$

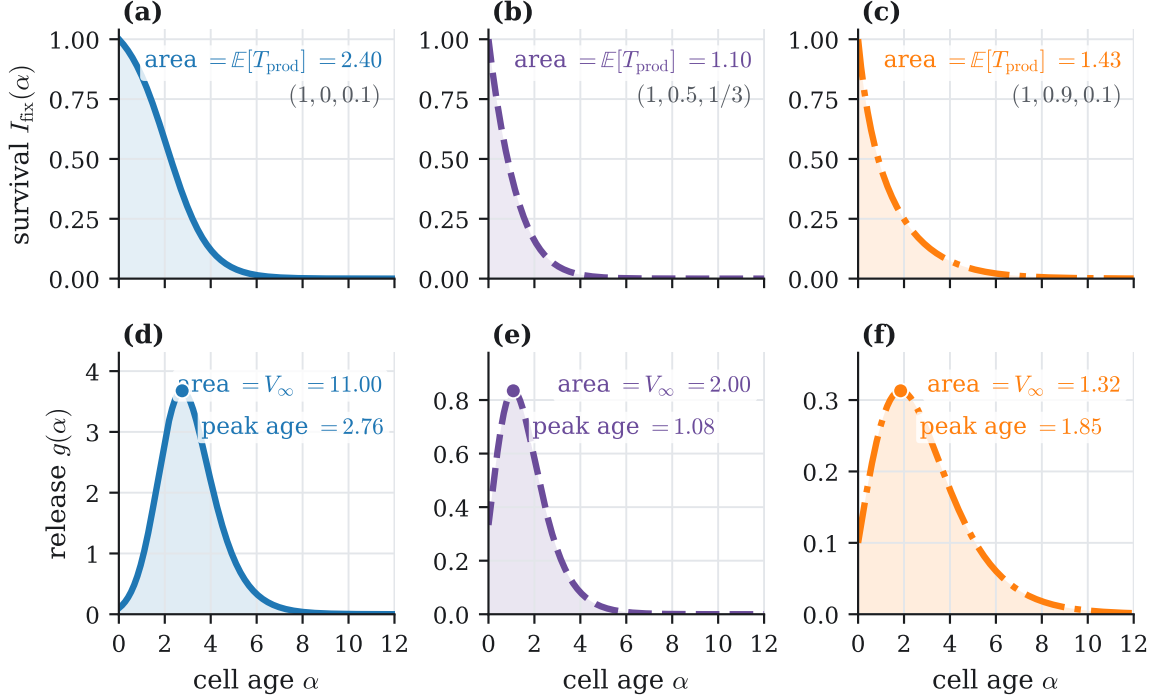


Figure 1.2: Survival I_{fix} (a)–(c) and release $g = \delta K$ (d)–(f) across the flooding boundary of [table 1.2](#). Release rises then falls — a cell must accumulate before it bursts — while survival decays; shaded areas check $\int_0^\infty I_{\text{fix}} = \mathbb{E}[T_{\text{prod}}]$ and $\int_0^\infty g = V_\infty$. Parameters, left to right: $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $(1, 0.5, 1/3)$, $(1, 0.9, 0.1)$, with $L = 1.10, 1.00, 0.42$ and $V_\infty = 11.00, 2.00, 1.32$. Independent g -scales keep the timing shift visible.

The reasoning behind (1.8) is the one the failed proposal of [section 1.2.3](#) stated and then did not use: a cell bursts at rate δX_α and releases X_α particles when it does, so the expected number released in $[\alpha, \alpha + d\alpha]$ is $\delta \mathbb{E}[X_\alpha^2] d\alpha = \delta K(\alpha) d\alpha$. With K from (1.54) both kernels are explicit functions of age, which is what it means for the population model to be *forced by* the intracellular process ([figure 1.2](#)).

1.3.2 The renewal system

Hold target cells T fixed, let infection occur at rate γ per virion, clear virions at rate c , and define the incidence

$$i(t) = \gamma T \mathcal{V}(t). \quad (1.9)$$

Each infection event at time $t - \alpha$ starts a cell of age α , still productive with probability $I_{\text{fix}}(\alpha)$ and releasing at expected rate $g(\alpha)$. Summing over infection times and adding the $\mathcal{I}_0$ cells already infected at $t = 0$ gives the system.

Definition 1.1 (The burst-aware renewal system).

$$\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + \int_0^t i(t - \alpha) I_{\text{fix}}(\alpha) d\alpha = \mathcal{I}_0 I_{\text{fix}} + (i * I_{\text{fix}})(t), \quad (1.10)$$

$$\frac{d\mathcal{V}}{dt}(t) = \mathcal{I}_0 g(t) + \int_0^t i(t - \alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g)(t) - c \mathcal{V}(t). \quad (1.11)$$

These are not an approximation to the underlying stochastic process. They are its first moments exactly, and the reason is that every rate in sight is linear.

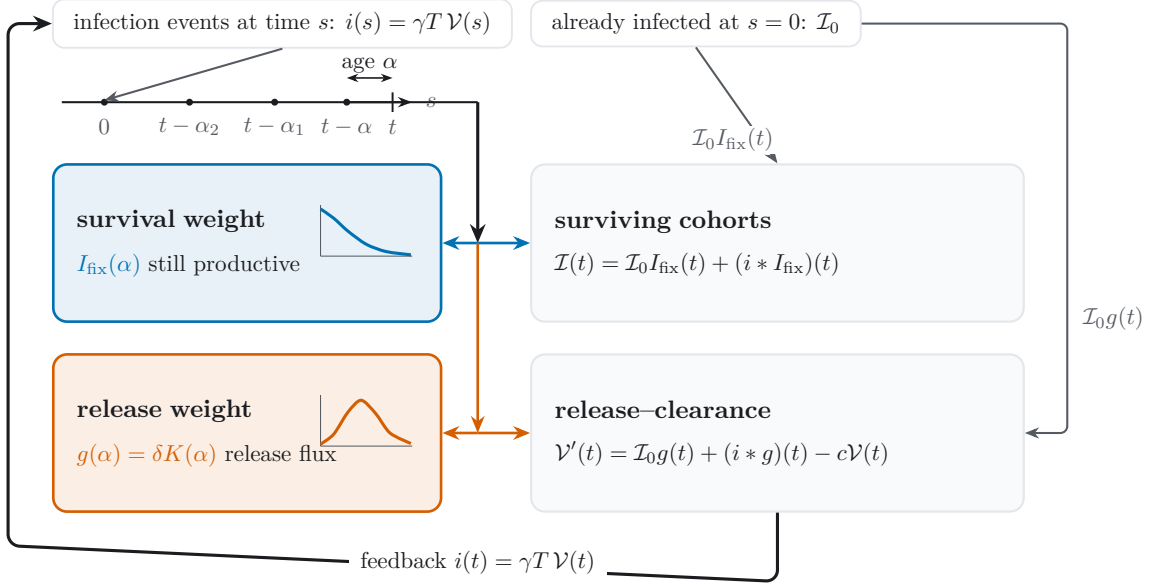


Figure 1.3: Bookkeeping of [definition 1.1](#): infection events start cohorts; the kernels I_{fix} and $g = \delta K$ weight them by age; the feedback $i = \gamma T V$ closes the loop.

Proposition 1.2 (The renewal system is exact in the mean). *Consider the stochastic process in which each infection event starts an independent birth–death–catastrophe cell, each released particle independently infects at rate γT or clears at rate c , and the $\mathcal{I}_0$ cells present at $t = 0$ have age zero. Write $\mathcal{I}(t)$ and $\mathcal{V}(t)$ for the expected numbers of productive cells and free particles. Then $\mathcal{I}$ and $\mathcal{V}$ satisfy [\(1.10\)](#) and [\(1.11\)](#) exactly.*

Proof. Let N be the point process of infection events. Since each free particle infects at rate γT and particles are independent, N has compensator $\gamma T \mathcal{V}^\omega(t) dt$, where $\mathcal{V}^\omega$ is the realised particle count; taking expectations, its mean intensity is $i(t) = \gamma T \mathcal{V}(t)$. A cell founded at time $t - \alpha$ evolves independently of N and of every other cell, so it is still productive at t with probability $I_{\text{fix}}(\alpha)$ and its expected release flux at t is $g(\alpha)$, both by definition of the two kernels. Conditioning on N and applying Campbell’s formula — legitimate because I_{fix} and g are bounded and non-negative, so Fubini applies — gives $\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + (i * I_{\text{fix}})(t)$ and an expected release flux of $\mathcal{I}_0 g(t) + (i * g)(t)$. Clearance removes particles at rate c per particle, contributing $-c \mathcal{V}(t)$ to the mean. Collecting the terms gives [\(1.11\)](#). $\square$

The step that would fail for a nonlinear incidence is the identification of the mean intensity with $\gamma T \mathcal{V}$: for $f(\mathcal{V}, T)$ nonlinear, $\mathbb{E}[f(\mathcal{V}^\omega)] \neq f(\mathbb{E}[\mathcal{V}^\omega])$ and the moment hierarchy does not close. Assumption (A1), which holds T fixed, is what keeps the incidence linear. Test H of [section 1.C](#) checks [proposition 1.2](#) against an independent Gillespie ensemble.

[Table 1.3](#) sets the two systems side by side. Both constants are replaced, not only the release: memoryless removal $d_{\mathcal{I}} \mathcal{I}$ becomes the age structure implicit in [\(1.10\)](#), and constant release $p \mathcal{I}$ becomes the convolution with g . A replacement touching only the release term — the first thought of [section 1.2.3](#) — would leave the removal side memoryless, and still wrong.

Differentiating [\(1.10\)](#) makes the structure visible:

$$\frac{d\mathcal{I}}{dt} = i(t) + \mathcal{I}_0 I'_{\text{fix}}(t) + \int_0^t i(t - \alpha) I'_{\text{fix}}(\alpha) d\alpha, \quad (1.12)$$

in which incidence appears as a source exactly as in the classical model, while the remaining terms encode age-dependent loss of productivity. There is in general no constant $d_{\mathcal{I}}$ reducing

Classical BMVR (1.1)	Renewal system (1.10)–(1.11)
$\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}$
constant removal rate $d_{\mathcal{I}}$	age-structured survival via I_{fix}
$\mathcal{V}' = p\mathcal{I} - c\mathcal{V}$	$\mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}$
constant release rate p	age-structured release via $g = \delta K$

Table 1.3: What the renewal system replaces. Both constants go, not only the release rate.

this to $\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ for every history of i : that holds only when everything grows exponentially, and the exception is what [section 1.4](#) exploits.

Remark 1.3 (What is and is not new). The renewal framework is standard — it is the McKendrick–von Foerster machinery of mathematical demography [4, 5, 6] — and so are age-structured within-host models: Nelson et al. [7] write down a within-host system with age-dependent production and death rates that is structurally the same as (1.10)–(1.11), and take the age dependence as given. What is new here is where the kernels come from. In (1.7) and (1.8) the survival and release schedules are *derived in closed form from a mechanistic intracellular stochastic process*, so that the population model has no free age structure to fit, and the consequences drawn in the next two sections — the effective-parameter map, the identifiability obstruction, the flooding criterion — follow from the intracellular rates alone.

1.3.3 Generality: any pair of kernels will do

Nothing in (1.10) and (1.11) used the birth–death–catastrophe process beyond the two kernels it supplied. With incidence $i(t) = \gamma T \mathcal{V}(t)$, or a nonlinear $f(\mathcal{V}, T)$, *any* pair (S, g) gives

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + (i * g)(t) - c\mathcal{V}(t), \quad (1.13)$$

with effective parameters $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c) \int_0^\infty g$. This is the chapter’s most general statement, and what makes release mechanisms commensurable: budding, bursting, multiply infected cells, recurrent partial dumps and multi-stage eclipse differ only in their (S, g) ([table 1.4](#)). The first row is the structural check of [figure 1.17](#), that exponential kernels reduce (1.13) exactly to classical BMVR; [section 1.6](#) works through the rest.

Model	$S(\alpha)$	$g(\alpha)$
Classical BMVR	$e^{-d_{\mathcal{I}}\alpha}$	$p e^{-d_{\mathcal{I}}\alpha}$
Absorbing BDC (this chapter)	$I_{\text{fix}}(\alpha)$	$\delta K(\alpha)$
MOI- k BDC (the chapter on its distribution theory)	$I(\alpha)^k - D(\alpha)^k$	$\delta K_k(\alpha)$
Reset catastrophe (section 1.6.2)	$e^{-d_{\mathcal{I}}\alpha}$	$\delta e^{-d_{\mathcal{I}}\alpha} \mathbb{E}[X_\alpha^2]$
HIV multi-stage (section 1.6.5)	stage survival	$p \times$ productive occupancy
Two-type / GATE (the two-type chapter)	two-type I_{fix}	two-type second-moment flux

Table 1.4: One skeleton, many mechanisms. Each row is a different single-cell story; all of them induce (1.13), and differ only in the pair of kernels they supply.

1.3.4 Numerical verification

The renewal system was checked end to end by a numerical suite at four levels: kernel integrity, the reduction at $\gamma T = 0$, the structural limits available in closed form, and stochastic ground

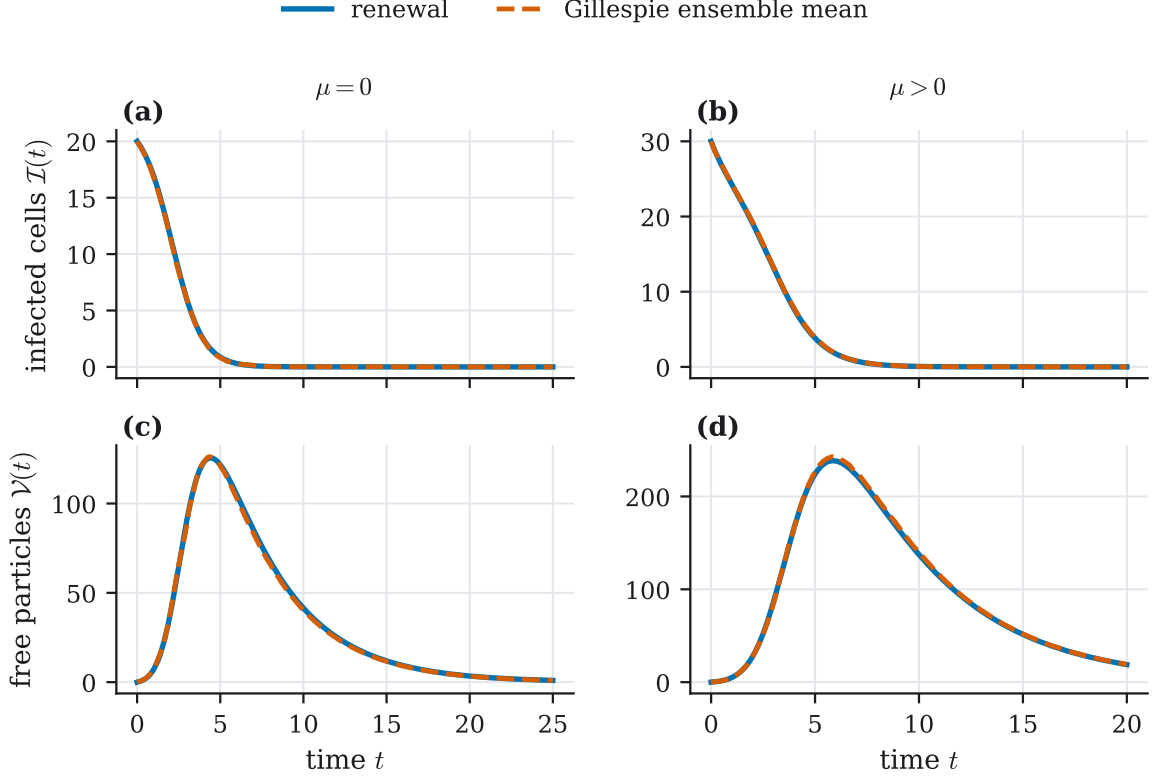


Figure 1.4: Stochastic ground truth. Ensemble means of an independent Gillespie simulation — birth–death–catastrophe cells with impulse releases at burst times, exponential clearance, no reinfection — against (1.10)–(1.11). Left column $\mu = 0$, right column $\mu > 0$; (a)–(b) infected cells, (c)–(d) free particles. The curves are indistinguishable at plotting resolution and the relative L^2 discrepancy of 1–2% is sampling error: the renewal equations are not an approximation to the simulation, and this is the check that says so.

truth. All 62 checks pass. Exponential kernels reproduce classical BMVR to relative error below 5×10^{-5} ; the late growth rate equals the characteristic root of section 1.4 to about 10^{-7} ; the R_0 threshold is sharp; and an independent Gillespie ensemble reproduces the renewal means for $\mathcal{I}$ and $\mathcal{V}$ to within 1–2% (figure 1.4). The catalogue is section 1.C.

1.4 The classical model as a projection

The renewal system does not merely disagree with classical BMVR in transients; it contains classical BMVR as an exact special case, and the relationship is precise enough to be turned into predictions.

1.4.1 Exact reduction in the exponential phase

Suppose the incidence grows exponentially, $i(t) = i_0 e^{rt}$, at times late enough that the lower limits of the convolutions are irrelevant. Then

$$\mathcal{I}(t) = i_0 e^{rt} \widetilde{I_{\text{fix}}}(r), \quad \mathcal{V}'(t) = i_0 e^{rt} \delta \widetilde{K}(r) - c \mathcal{V}(t), \quad (1.14)$$

where $\widetilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha) d\alpha$. A classical model driven by the same incidence has $\mathcal{I}(t) = i_0 e^{rt} / (r + d_{\mathcal{I}})$ and the eigenvalue relation $(r + d_{\mathcal{I}})(r + c) = \gamma T p$; matching the two exponential

solutions fixes

$$p_{\text{eff}}(r) = \frac{\delta \widetilde{K}(r)}{\widetilde{I}_{\text{fix}}(r)}, \quad d_{\mathcal{I},\text{eff}}(r) = \frac{1}{\widetilde{I}_{\text{fix}}(r)} - r. \quad (1.15)$$

The reduction is exact, and it is the central claim of the chapter.

Theorem 1.4 (Exponential-phase equivalence). *Fix a growth rate $r > -\theta$. The constant pair $(p_{\text{eff}}(r), d_{\mathcal{I},\text{eff}}(r))$ of (1.15) is the unique one for which classical BMVR reproduces the renewal system's e^{rt} solution exactly, in both $\mathcal{I}$ and $\mathcal{V}$; and substituting it into the classical eigenvalue relation returns*

$$r + c = \gamma T \delta \widetilde{K}(r), \quad (1.16)$$

precisely the characteristic equation of the renewal system. What no constant pair reproduces is a transient, which samples a continuum of growth rates at once.

Proof. Uniqueness and existence are (1.14) read as two equations in two unknowns: matching $\mathcal{I}$ forces $1/(r + d_{\mathcal{I}}) = \widetilde{I}_{\text{fix}}(r)$, which is $d_{\mathcal{I},\text{eff}}$, and matching the release term then forces $p \widetilde{I}_{\text{fix}}(r) = \delta \widetilde{K}(r)$, which is p_{eff} . Both transforms are finite and strictly positive for $r > -\theta$, so the pair is well defined. Substituting into $(r + d_{\mathcal{I}})(r + c) = \gamma T p$ gives $(r + c)/\widetilde{I}_{\text{fix}}(r) = \gamma T \delta \widetilde{K}(r)/\widetilde{I}_{\text{fix}}(r)$, that is (1.16). That no single pair serves a transient follows from proposition 1.6 below: p_{eff} is not constant in r . $\square$

So the classical model is not a rival to the renewal system but a projection of it, exact on one exponential phase at a time. Everything else in this section is a consequence of reading (1.15) as a map.

The two ends of the map are explicit. Using $\int_0^\infty \delta K = V_\infty$ (from $V' = \delta K$) and $\int_0^\infty I_{\text{fix}} = \mathbb{E}[T_{\text{prod}}]$ ((1.57)),

$$r = 0 : \quad p_{\text{eff}}(0) = \frac{V_\infty}{\mathbb{E}[T_{\text{prod}}]} = \frac{\lambda V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I},\text{eff}}(0) = \frac{\lambda}{\log(a/(a-1))}; \quad (1.17)$$

the $r = 0$ value coincides with naive moment-matching, lifetime output divided by lifetime — a consistency check, and the matching used in the overlays below. At the other end,

$$r \rightarrow \infty : \quad p_{\text{eff}}(r) \rightarrow \delta, \quad d_{\mathcal{I},\text{eff}}(r) \rightarrow \mu + \delta, \quad (1.18)$$

the *young-cell* limit: in a fast-growing epidemic almost every infected cell was infected moments ago, holds one particle, bursts at rate δ , and releases one.

Calling these the two ends understates the map, because $r = 0$ is not an end. Both transforms converge for every $r > -\theta$, and negative r is not a fiction: it is a decaying infection, the case of figure 1.6(b,d). The genuine left-hand endpoint is the mirror of the young-cell limit.

Proposition 1.5 (The old-cell limit). *The release flux per surviving cell tends to a finite limit,*

$$\lim_{\alpha \rightarrow \infty} \frac{g(\alpha)}{S(\alpha)} = \frac{a[2\lambda(1-b) + \delta]}{a-1} = \frac{\delta a(a+1)}{(a-1)^2} = \delta \mathbb{E}[X^2]_{\text{QS}}, \quad (1.19)$$

the burst rate times the second moment of the quasi-stationary load, and consequently $p_{\text{eff}}(r) \rightarrow \delta \mathbb{E}[X^2]_{\text{QS}}$ as $r \downarrow -\theta$.

Proof. Write $v = e^{-\theta\alpha}$. As $\alpha \rightarrow \infty$, $v \rightarrow 0$ and (1.52) and (1.53) give $I_{\text{fix}} \sim (a-b)^2 v / (Aa)$ and $I \rightarrow b$, so by (1.54) $g = 2\lambda(\kappa - I)J \sim 2\lambda(\kappa - b)(a-b)^2 v / A^2$. The ratio of the two is $2\lambda(\kappa - b)a/A$, which is the first equality after $2\lambda(\kappa - b) = 2\lambda(1-b) + \delta$. For the second, the identity $AB = \delta/\lambda$ of (1.51) reads $\lambda(1-b) = \delta/(a-1)$, whence $a[2\delta/(a-1) + \delta]/(a-1) = \delta a(a+1)/(a-1)^2$. That last expression is $\delta \mathbb{E}[X^2]_{\text{QS}}$ because the quasi-stationary load is geometric on $\{1, 2, \dots\}$ with

ratio $1/a$ ((1.59)), whose second moment is $a(a+1)/(a-1)^2$. Finally, both $\widetilde{I}_{\text{fix}}$ and $\delta\widetilde{K}$ diverge like $1/(r+\theta)$ as $r \downarrow -\theta$, with coefficients in the ratio just computed, so their quotient tends to it. $\square$

Two things are worth saying about (1.19). It repairs the failed proposal of section 1.2.3, which offered the quasi-stationary *mean* $\mathbb{E}[X]_{\text{QS}}$ as a release rate and was wrong twice over: a count was used where a rate was needed, and the first moment where the second is called for. Both corrections appear in (1.19) together — δ supplies the missing inverse time, and the load-proportional burst hazard weights by X a second time. The naive proposal was, in that sense, an old-cell limit with the wrong moment and no clock. And it gives the absorbing process the mature-cell limit that section 1.6.2 gives the resetting one, in the same form $p_\infty = \delta \mathbb{E}_\pi[X^2]$ with π the quasi-stationary law.

Between the endpoints the map is monotone, and the reason is a single property of the kernels.

Proposition 1.6 (Monotonicity of the effective release rate). *Write $\rho(\alpha) = g(\alpha)/S(\alpha)$ for the expected release flux per still-productive cell, so that $\rho(0) = \delta$ and $\rho(\infty) = \delta \mathbb{E}[X^2]_{\text{QS}}$. If ρ is non-decreasing then p_{eff} is non-increasing on $(-\theta, \infty)$, strictly so if ρ is strictly increasing.*

Proof. For $r > -\theta$ let $\pi_{f,r}$ be the probability density $\alpha \mapsto e^{-r\alpha} f(\alpha)/\widetilde{f}(r)$. Differentiating under the integral, $d \log \widetilde{f}(r)/dr = -\mathbb{E}[\alpha]_{\pi_{f,r}}$, so

$$\frac{d}{dr} \log p_{\text{eff}}(r) = \mathbb{E}[\alpha]_{\pi_{S,r}} - \mathbb{E}[\alpha]_{\pi_{g,r}}. \quad (1.20)$$

The two densities have ratio $\pi_{g,r}(\alpha)/\pi_{S,r}(\alpha) \propto \rho(\alpha)$, independently of r . If ρ is non-decreasing, $\pi_{g,r}$ dominates $\pi_{S,r}$ in the monotone likelihood ratio order, hence in the usual stochastic order, hence in mean; so (1.20) is non-positive. $\square$

The hypothesis is a statement about one closed-form function of one variable: substituting $v = e^{-\theta\alpha}$ into (1.52)–(1.54) gives

$$\rho = \frac{2\lambda [B(\kappa - a)v + A(\kappa - b)](a - bv)}{(Bv + A)^2}, \quad (1.21)$$

which must be shown decreasing in $v \in (0, 1]$. That reduces the question to a finite algebraic one; it is not settled here, but ρ was found increasing at every one of 4000 triples sampled over $(\mu, \delta) \in [10^{-3}, 20]^2$ at $\lambda = 1$, and p_{eff} decreasing at every one (test M of section 1.C).

The size of the effect is what matters. At $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $p_{\text{eff}}(0) \approx 4.59$ against $p_{\text{eff}}(\infty) = 0.1$ — the same cell, two orders of magnitude apart, with nothing separating the two figures but how fast the epidemic around it is growing. Over the whole map the span is larger, and exact: by (1.18) and (1.19) the ratio of the two endpoints is

$$\frac{\delta \mathbb{E}[X^2]_{\text{QS}}}{\delta} = \mathbb{E}[X^2]_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}, \quad (1.22)$$

a pure number fixed by the intracellular rates alone. It is 231 at $(1, 0, 0.1)$ and 30.5 at $(1, 0.9, 0.1)$: the full dynamic range of a fitted release rate is the second moment of the quasi-stationary load.

The closed hypergeometric forms of $\widetilde{I}_{\text{fix}}$ and $\delta\widetilde{K}$ are in section 1.D.1; the curves above are computed by direct quadrature.

1.4.2 The generation kernel

One object lies behind both (1.15) and the comparison of section 1.5. Define the cell-to-cell *generation kernel*

$$\mathcal{A}(\alpha) = \gamma T (g * e^{-c \cdot})(\alpha) = \gamma T \int_0^\alpha g(u) e^{-c(\alpha-u)} du, \quad (1.23)$$

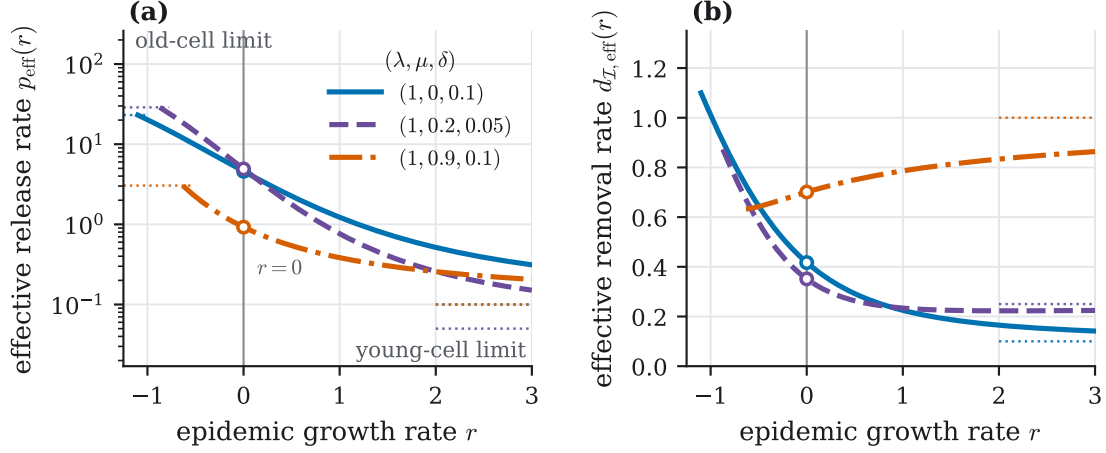


Figure 1.5: The effective release rate (a) and removal rate (b) of (1.15) against the epidemic growth rate r , for three parameter sets, over the whole domain $r > -\theta$ on which the map is defined. Open circles mark the $r = 0$ values (1.17); dotted segments mark the limits each curve runs toward. Note that $r = 0$ is an interior point, not an end: the left-hand endpoint is the old-cell limit $\delta \mathbb{E}_{\text{QS}}[X^2]$ and the right-hand one the young-cell limit δ . Read the vertical extent of each curve as the size of the error made by treating p as a constant of the pathogen rather than of the epidemic's phase.

the rate at which a cell of age α generates new infected cells: it releases at rate $g(u)$ at age u , a particle released then is still free a time $\alpha - u$ later with probability $e^{-c(\alpha-u)}$, and infects at rate γT . Its transform is

$$\tilde{\mathcal{A}}(r) = \frac{\gamma T \tilde{g}(r)}{r + c}, \quad (1.24)$$

so the characteristic equation (1.16) is exactly $\tilde{\mathcal{A}}(r) = 1$: the Euler–Lotka equation of a one-type age-structured population, with the infected cell as the individual and $\mathcal{A}$ as its mean reproduction measure [8, 6].

Three things follow, each of which the chapter already contains in another form. First, the total mass of $\mathcal{A}$ is the basic reproduction number,

$$R_0 = \int_0^\infty \mathcal{A}(\alpha) d\alpha = \gamma T \int_0^\infty g \cdot \int_0^\infty e^{-cu} du = \frac{\gamma T V_\infty}{c}, \quad (1.25)$$

which depends on g only through its integral. Second, the normalised kernel $\mathcal{A}(\alpha)/R_0$ is the *generation-time density*, the law of the age at which a cell produces a randomly chosen one of its offspring cells; so the mechanism named in section 1.5.5 is a definition rather than an interpretation, and the numbers that go with it follow from g alone. By (1.23) the mean generation time is the mean release age under g/V_∞ plus the mean extracellular wait $1/c$: for the budding comparator matched at $r = 0$ the release age is exponential with mean $\mathbb{E}[T_{\text{prod}}] = 2.398$, giving $2.398 + 1 = 3.398$ at $c = 1$; for the bursting cell it is 3.089, giving $3.089 + 1 = 4.089$. Third, since r solves $\tilde{\mathcal{A}}(r) = 1$ while $R_0 = \tilde{\mathcal{A}}(0)$, threshold and growth rate are different functionals of one kernel — the standard R_0 -versus- r distinction [9], here with both sides in closed form.

1.4.3 What the projection loses

To see what age structure does to an epidemic, integrate both systems side by side. In figures 1.6 and 1.7 the solid curve is the renewal system and the dashed one is classical BMVR with the fairest constant match, at the $r = 0$ end of (1.15),

$$p = p_{\text{eff}}(0) = \frac{V_\infty}{\mathbb{E}[T_{\text{prod}}]} = \frac{\lambda V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I}} = d_{\mathcal{I},\text{eff}}(0) = \frac{\lambda}{\log(a/(a-1))}, \quad (1.26)$$

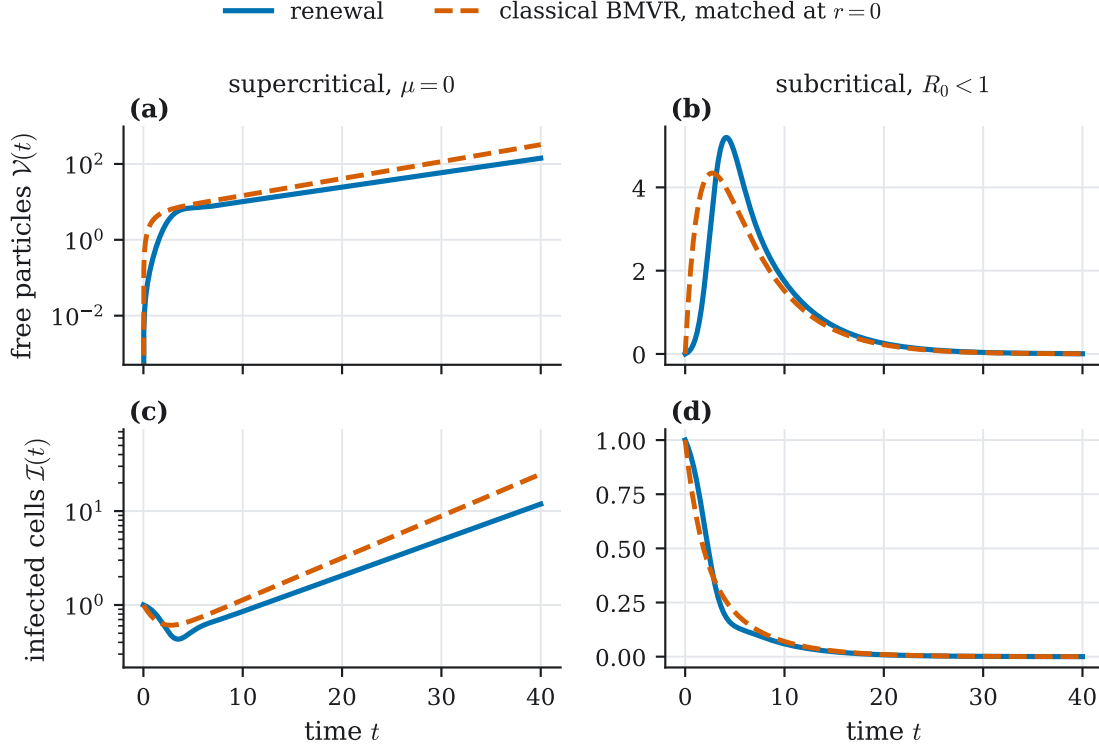


Figure 1.6: Both state variables, both sides of the threshold. Free particles $\mathcal{V}(t)$ (top row) and infected cells $\mathcal{I}(t)$ (bottom) under the renewal system (solid) and under classical BMVR matched at $r = 0$ (dashed), for a supercritical infection (left, $R_0 = 1.76$) and a subcritical one (right, $R_0 = 0.28$). The matched model gets the order of magnitude and not the shape; and the discrepancy is not confined to the release equation — age structure moves the cell trajectory too, which is why matching p alone cannot repair it. Parameters: $(\lambda, \mu, \delta) = (1, 0, 0.1)$ throughout, with $\gamma T = 0.04$, $c = 0.25$ on the left and $\gamma T = 0.01$, $c = 0.4$ on the right.

so that total lifetime output and mean productive lifetime agree. The comparator is the budding cell of [section 1.2.1](#), its constants now chosen by the map rather than by hand.

The reading is consistent across panels. The matched model misses three things: the early lag, because young cells dominate and release is nearer the young-cell limit $p \rightarrow \delta$ than $p_{\text{eff}}(0)$; the timing of the peak; and the late growth rate, since generation-time structure moves the operative release rate away from its $r = 0$ value as the epidemic grows. The relative free-particle discrepancy,

$$\frac{|\mathcal{V}^{\text{ren}}(t) - \mathcal{V}^{\text{cl}}(t)|}{\max_s \mathcal{V}^{\text{ren}}(s)}, \quad (1.27)$$

is often order one or larger by the end of the plotted window in supercritical regimes, and worst when intracellular death is high, where V_∞ is small while the age structure of release stays strongly non-exponential. Discrepancies grow with R_0 and with that age structure — slow bursting, non-zero μ .

These are the facts the reduction predicts: a constant pair reproduces one exponential growth rate exactly, and a transient sampling a continuum of ages is beyond it.

1.4.4 The invariance of R_0

Setting $r = 0$ in (1.16) gives $\gamma T \delta \widetilde{K}(0) = c$, that is $\gamma T V_\infty = c$, so the renewal system's basic reproduction number is

$$R_0^{\text{ODE}} = \frac{\gamma T V_\infty}{c}. \quad (1.28)$$

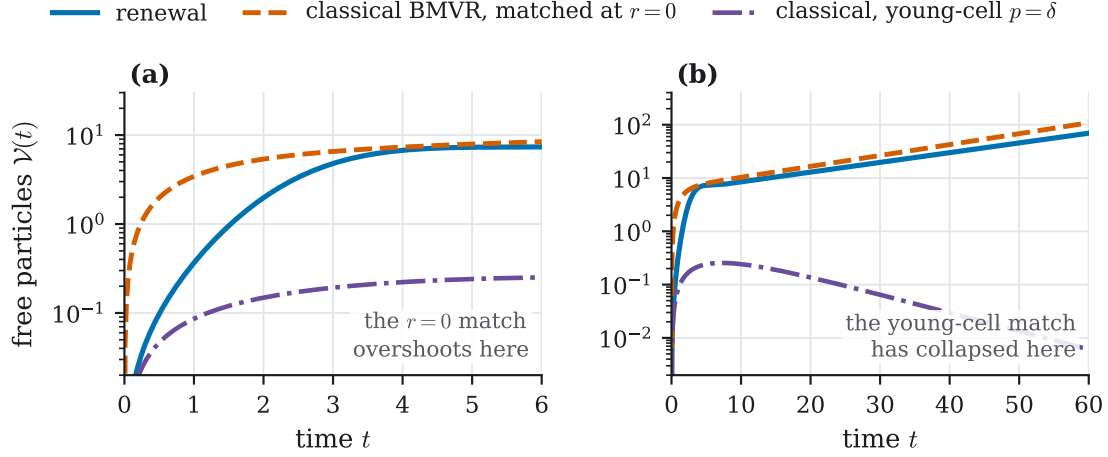


Figure 1.7: Why no single constant pair will do. One renewal trajectory (solid) against two classical parameter choices: the $r=0$ match (dashed) and the young-cell limit $p=\delta$, $d_{\mathcal{I}} = \mu + \delta$ (dash-dot). (a) The first six time units, where the $r=0$ match overshoots by more than an order of magnitude while the young-cell parameters are the closer of the two. (b) The full window, where the ordering has reversed and the young-cell model has collapsed. Each choice is correct at one growth rate and the trajectory passes through many, exactly as (1.15) predicts. Both panels logarithmic in $\mathcal{V}$. Parameters: $(\lambda, \mu, \delta) = (1, 0, 0.1)$, $\gamma T = 0.025$, $c = 0.2$, $R_0 = 1.38$.

This agrees with (1.25) [8].

Theorem 1.7 (R_0 is unchanged by bursting). *At fixed total per-cell output V_∞ , replacing constant budding by bursting leaves R_0 exactly unchanged. Bursting alters the generation-time distribution — hence the growth rate r , hence the extinction probability, hence the stochastic early phase — but not the threshold itself.*

Proof. By (1.25), $R_0 = \int_0^\infty \mathcal{A}$ depends on the release kernel only through $\int_0^\infty g = V_\infty$. Replacing bursting by budding at matched yield replaces g by another kernel with the same integral, and leaves R_0 where it was. The generation-time density $\mathcal{A}/R_0$, by contrast, depends on the whole of g , and it is through that density that r is determined by $\tilde{\mathcal{A}}(r) = 1$. $\square$

Two remarks attach to the theorem. The first is evidential: the numerical checks of figure 1.18 put the late growth rate at the characteristic root r_* of (1.16) to about 10^{-7} across five scenarios, and trajectories decay for $R_0 < 1$ and grow for $R_0 > 1$, with the transition resolved already at $R_0 = 1.05$. The second is a caution about assumption (A2): (1.28) is the *continuum* reproduction number, in which one virion contributes mean $\gamma T/c$ new infections, whereas a discrete branching process absorbing virions on infection carries the mean $m = qV_\infty$ with $q = \gamma T/(\gamma T + c)$. The two agree when $\gamma T \ll c$, and it is m that governs section 1.5.

1.4.5 Identifiability: a negative result

How much of the intracellular process can a population-level fit recover? At any fixed growth rate r , the map

$$(\lambda, \mu, \delta) \longmapsto (p_{\text{eff}}(r), d_{\mathcal{I}, \text{eff}}(r)) \quad (1.29)$$

sends three rates to two numbers, and is therefore not invertible (figure 1.8). This is a structural identifiability statement of the kind Miao et al. [10] treat for viral dynamics generally, and it is negative for the same reason: the observable is a projection, and the projection has a kernel. It also closes the ambition of section 1.2.3, which hoped that a good estimate of p would give access to the rates making up the quasi-stationary mean: it does not, and neither does $(p, d_{\mathcal{I}})$ together.

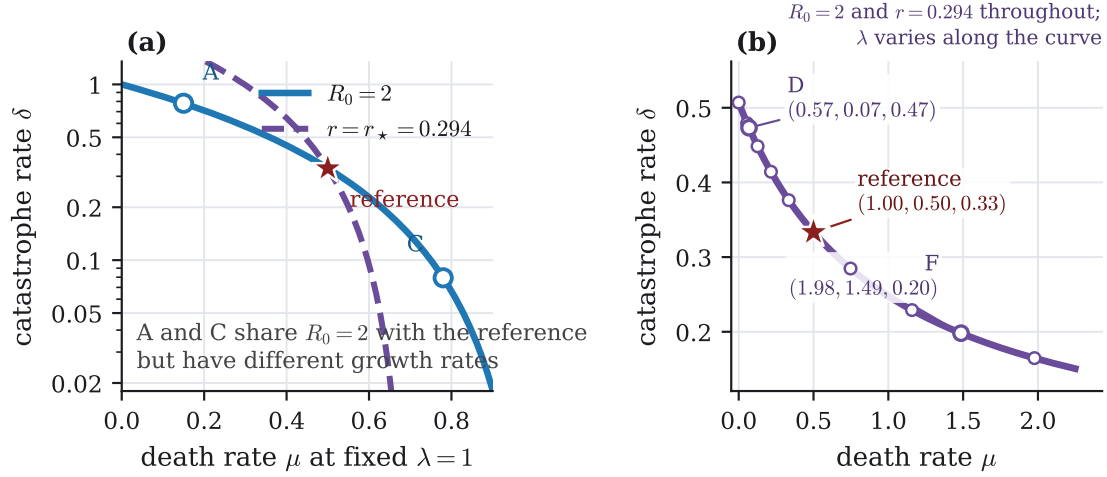


Figure 1.8: Population observables leave level sets rather than identifying the intracellular rates. With $c = 1$ and $\gamma T = 1$, the interior flooding point $(\lambda, \mu, \delta) = (1, 0.5, 1/3)$ has $R_0 = 2$ and $r_* = 0.294$. (a) One population observable leaves a level curve: in the slice $\lambda = 1$ the contours $R_0 = 2$ and $r = r_*$ are distinct; alternatives A and C share R_0 with the reference but not its growth rate. (b) With λ free, a continuous family of triples shares both. Joint knowledge of threshold and exponential growth therefore does not identify (λ, μ, δ) .

A third observable is required; in order of experimental accessibility:

1. the burst-size dispersion — the geometric ratio $1/a$ fixes one combination of the rates directly (the burst-size law of the chapter on its distribution theory);
2. the fraction of cells that clear without bursting, which is b ;
3. the shape — not merely the mean — of the release-time density δJ (the chapter on its distribution theory).

Any one of these, with (p, d_T) , over-determines (λ, μ, δ) and converts the fit into a consistency test of the intracellular model. The limitation is not one of fitting procedure but of the information content of classical BMVR itself.

1.4.6 A prediction for fitting practice

The fitting consequences — acute-phase versus set-point estimates of p , and the eclipse-like lag visible in figures 1.6 and 1.7 — are taken up in section 1.7.1. The lag *is* the eclipse: it emerges from the intracellular dynamics with no extra parameter.

1.5 Bursting versus budding

Does bursting help a pathogen get established? The question belongs to the stochastic early phase, where counts are small and extinction is a live possibility [11], and here it can be answered exactly: both versions of the early infection are branching processes whose offspring distributions are in closed form.

The mechanism at issue is worth naming before it is analysed. A bursting cell delivers its whole output in one instant, so a founder that survives to rupture presents the target population with many simultaneous infection attempts rather than a trickle spread over its lifetime. Whether that concentration helps or hurts is not obvious: many attempts at once buy insurance against

the loss of any one particle, but they are all forfeited if the intracellular population dies before the burst. We call the concentration *flooding* throughout, and the question of this section is when it pays.

1.5.1 Offspring of one infected cell

In the branching approximation each released particle independently either infects a target cell, at rate γT , or is cleared, at rate c , so infects with probability

$$q = \frac{\gamma T}{\gamma T + c}. \quad (1.30)$$

The offspring of one infected cell is the burst size $\mathcal{K}$ thinned by that probability. Summing (1.59), the unconditional burst-size probability generating function is $G_{\mathcal{K}}(z) = b + (\delta/\lambda) z/(a - z)$ — including the mass b of cells that never burst — and thinning replaces z by $y := 1 - q + qz$:

$$G_{\text{off}}(z) = b + \frac{\delta}{\lambda} \frac{y}{a - y}, \quad y = 1 - q + qz. \quad (1.31)$$

One checks $G_{\text{off}}(1) = b + (\delta/\lambda)/(a - 1) = b + B = 1$. The mean and variance of the offspring number are

$$m = G'_{\text{off}}(1) = q V_{\infty}, \quad \text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}}{a - 1} + q V_{\infty} - q^2 V_{\infty}^2, \quad (1.32)$$

the variance following from $\text{Var}(\mathcal{K})$ via the limit of (1.61) and the composition formula for random sums. When $\gamma T \ll c$, $q \approx \gamma T/c$ and $m \approx R_0^{\text{ODE}}$.

1.5.2 Extinction probability in closed form

The extinction probability of a branching process is the smallest root of $G_{\text{off}}(z) = z$ in $[0, 1]$. With $y = 1 - q + qz$ the fixed-point equation is quadratic, $z = 1$ is always a root, and the other root gives, for $m > 1$,

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q} = \frac{a - 1}{q} + 1 - a(1 - b). \quad (1.33)$$

1.5.3 Comparison with matched-mean budding

The comparator is the budding cell of section 1.2.1, whose output is geometric with mean $p/d_{\mathcal{I}}$. Match it to the bursting cell's total output, $p/d_{\mathcal{I}} = V_{\infty}$. That matching is what makes the comparison fair, and by theorem 1.7 it costs nothing: at equal total output the two mechanisms have exactly the same R_0 , so everything that follows compares two pathogens at one threshold, and any difference between them is a difference in how the output is scheduled rather than in how much of it there is. Thinning a geometric law leaves a geometric law, so the budding offspring number is geometric with the same mean $m = qV_{\infty}$, and its extinction probability is

$$z_{\text{ext}}^{\text{bud}} = \frac{1}{m} \quad (m > 1). \quad (1.34)$$

Write $L := a(1 - b) = a - \mu/\lambda$, so that $V_{\infty} = L/(a - 1)$.

Theorem 1.8 (The flooding advantage). *For $m > 1$,*

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right). \quad (1.35)$$

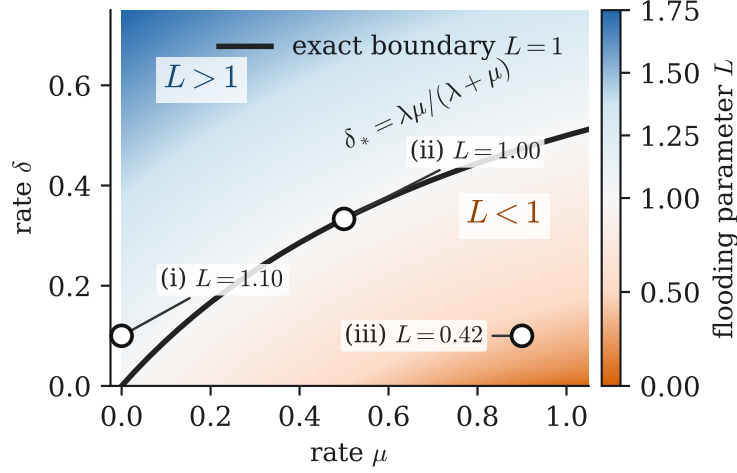


Figure 1.9: Where a pathogen sits relative to the flooding boundary. The flooding parameter $L(\mu, \delta)$ at $\lambda = 1$; the exact curve $L = 1$, equivalently $\delta_* = \lambda\mu/(\lambda + \mu)$, partitions the rate plane into a region where bursting lowers extinction risk and one where it raises it. The three marked points are the regimes of table 1.2, $(\mu, \delta) = (0, 0.1)$, $(0.5, 1/3)$ and $(0.9, 0.1)$, with $L = 1.10$, 1.00 and 0.42 .

Hence bursting strictly lowers the establishment extinction probability, relative to matched-mean budding, if and only if $L > 1$, i.e.

$$\delta(\lambda + \mu) > \lambda\mu \quad \Longleftrightarrow \quad \boxed{\frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu}} \quad (1.36)$$

(with the convention that the condition holds for all $\delta > 0$ when $\mu = 0$). When $L = 1$ the two extinction probabilities are identical for every q .

Proof sketch. The identity (1.35) is algebra: substitute (1.33), $m = qV_\infty = qL/(a - 1)$ and simplify. For the criterion, $L > 1$ is $a(1 - b) > 1$, i.e. $a > 1 + \mu/\lambda$; inserting the definition of a and squaring (both sides are positive, or the right-hand side is not) yields $(\lambda + \mu + \delta)^2 - 4\lambda\mu > 4\lambda^2(1 + \mu/\lambda - \eta)^2$, which reduces to $\delta(\lambda + \mu) > \lambda\mu$. $\square$

The identity is exact in each of the three regimes, and at both values of q tested on the boundary (table 1.5 and figure 1.10).

(λ, μ, δ)	q	$z_{\text{ext}}^{\text{burst}}$	$z_{\text{ext}}^{\text{bud}}$	verdict
$(1, 0, 0.1)$	0.2	0.400	0.455	bursting wins ($L = 1.10$)
$(1, 0.9, 0.1)$	0.9	0.935	0.844	budding wins ($L = 0.42$)
$(1, 0.5, 1/3)$	0.6	0.833	0.833	tie ($L = 1$)
$(1, 0.5, 1/3)$	0.9	0.556	0.556	tie ($L = 1$)

Table 1.5: Extinction probabilities from (1.33) and (1.34), and the sign of their difference. The gap is $(L - 1)(1/m - 1)$ in every row.

For $m \leq 1$ both processes die out almost surely and the comparison is empty; (1.35) changes sign there, an artefact of both formulae running outside their domain $[0, 1)$, not a reversal of the biology.

The flooding advantage is real, but it is not free. Bursting concentrates output into one batch, lowering the offspring variance relative to a budding cell's geometric output, but it carries the all-or-nothing hazard that the intracellular population dies out first — with probability b —

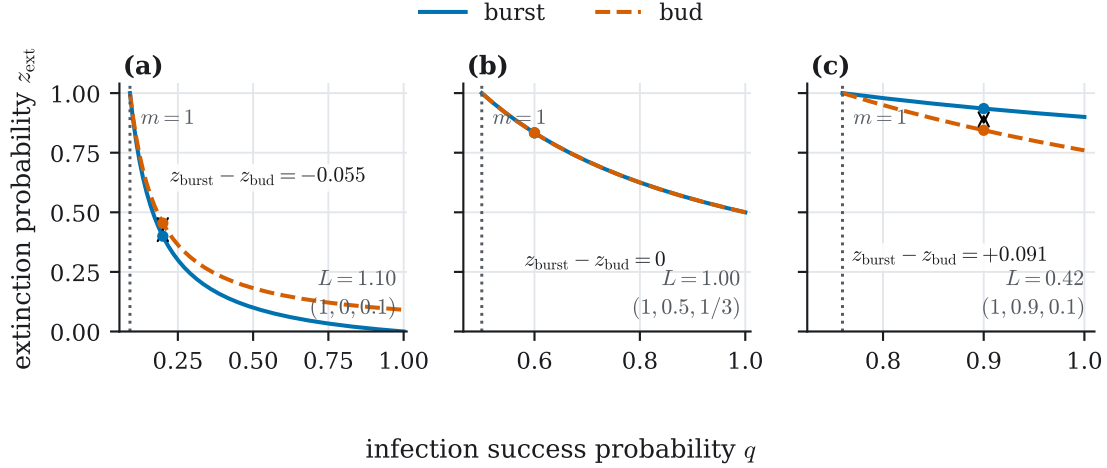


Figure 1.10: Extinction probabilities for bursting and matched-mean budding across the three regimes of table 1.2. Each pair of curves begins at $m = 1$; the signed gap follows (1.35). The q -windows differ because $m = 1$ sits at different q in each regime.

and the cell releases nothing. The criterion says the advantage survives exactly when the burst timescale $1/\delta$ is shorter than the harmonic-type combination $1/(1/\lambda + 1/\mu)$ of the birth and death timescales; when $\mu = 0$ the hazard vanishes and bursting always wins.

1.5.4 The boundary $L = 1$: a coincidence of laws

The boundary case is more than a tie of extinction probabilities.

Proposition 1.9. *The following are equivalent:*

$$L = 1 \iff a = 1 + \frac{\mu}{\lambda} \iff \delta(\lambda + \mu) = \lambda\mu \iff \frac{1}{\delta} = \frac{1}{\lambda} + \frac{1}{\mu}. \quad (1.37)$$

When they hold, the bursting offspring law (1.31) is exactly geometric with mean m — not merely equal in extinction probability to the budding comparator. In particular the offspring variances coincide, and

$$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}} = \frac{2q^2 V_\infty (L - 1)}{a - 1}, \quad (1.38)$$

so that at matched mean m the bursting offspring variance is smaller than the budding one when $L > 1$, equal when $L = 1$, and larger when $L < 1$.

Proof. $L = 1$ is $a(1 - b) = 1$, i.e. $a - 1 = ab = \mu/\lambda$, the chain of equivalences above. Under it, $b = \delta/\lambda$ (by (1.51)), and (1.31) collapses:

$$G_{\text{off}}(z) = b \left(1 + \frac{y}{a - y} \right) = \frac{ab}{a - y} = \frac{a - 1}{a - y} = \frac{(a - 1)/a}{1 - y/a},$$

using $ab = a - 1$. With $y = 1 - q + qz$,

$$1 - \frac{y}{a} = \frac{a - 1 + q}{a} \left(1 - \frac{q}{a - 1 + q} z \right),$$

so

$$G_{\text{off}}(z) = \frac{1 - \varrho}{1 - \varrho z}, \quad \varrho = \frac{q}{a - 1 + q} = \frac{m}{1 + m}, \quad m = \frac{q}{a - 1} = qV_\infty,$$

which is the probability generating function of a geometric law on $\{0, 1, 2, \dots\}$ with mean m — the same law as the matched budding comparator. The variance identity (1.38) is the difference between the geometric variance $m(1 + m)$ and (1.32), with $V_\infty = L/(a - 1)$. $\square$

At the boundary, then, bursting and budding are the same branching process in different clothes; away from it (1.38) quantifies how they differ.

1.5.5 The price: deterministic growth rate

Establishment is half the story; the other half is how fast an established infection grows. Match the models a second time, at R_0 and at mean productive lifetime: take the budding cell with $d_{\mathcal{I}} = 1/\mathbb{E}[T_{\text{prod}}]$ and $p = V_{\infty}d_{\mathcal{I}}$ — exactly $p_{\text{eff}}(0)$, so the fairest classical match available — and choose γT , or c , so both share $R_0 = 2$. Solving for the growth rates — r_{bud} from $(r + d_{\mathcal{I}})(r + c) = \gamma T p$, r_{burst} from (1.16) — gives the same ordering in all three regimes of table 1.2:

$$r_{\text{bud}} > r_{\text{burst}}. \quad (1.39)$$

The values, with $c = 1$, are table 1.6 and figure 1.11.

Whether (1.39) holds in general is not settled here, but it reduces to a single inequality, in which the coupling constants do not appear at all.

Proposition 1.10 (The ordering is a Laplace-transform comparison). *Match the comparator as above, so that $\int_0^{\infty} g_{\text{bud}} = \int_0^{\infty} g_{\text{burst}} = V_{\infty}$. Then $r_{\text{bud}} > r_{\text{burst}}$ for every pair $(\gamma T, c)$ with $R_0 > 1$ if and only if*

$$\tilde{g}_{\text{burst}}(r) \leq \tilde{g}_{\text{bud}}(r) = \frac{p}{r + d_{\mathcal{I}}} \quad \text{for all } r > 0, \quad (1.40)$$

that is, if and only if the release age under bursting dominates the matched exponential release age in the Laplace-transform order.

Proof. By (1.24) each model's growth rate is the unique root of $\phi(r) := \gamma T \tilde{g}(r) - (r + c)$; ϕ is strictly decreasing, and $\phi(0) = \gamma T V_{\infty} - c = c(R_0 - 1) > 0$, so the root is positive and unique. If (1.40) holds then $\phi_{\text{burst}} \leq \phi_{\text{bud}}$ pointwise, so the root of the former is no larger. Conversely, suppose $\tilde{g}_{\text{burst}}(r_0) > \tilde{g}_{\text{bud}}(r_0)$ for some $r_0 > 0$. Choose any $c > 0$ and set $\gamma T = (r_0 + c)/\tilde{g}_{\text{burst}}(r_0)$, so that $r_{\text{burst}} = r_0$; then $\phi_{\text{bud}}(r_0) < 0$ and $r_{\text{bud}} < r_0$. Taking c small makes $R_0 = \gamma T V_{\infty}/c$ exceed one, so the ordering fails for an admissible coupling. $\square$

Both sides of (1.40) agree at $r = 0$, by the matching, so its content is entirely in how the two release schedules are spread over age. The inequality was checked at 1500 triples drawn over $(\mu, \delta) \in [10^{-3}, 10] \times [10^{-3}, 16]$ at $\lambda = 1$, at 60 values of r across four decades, without a violation (test N of section 1.C). The *usual* stochastic order does not hold — the burst release-age survival function dips below the matched exponential — so (1.40) is not a weaker way of saying something stronger. It is the order the problem actually has.

(λ, μ, δ)	L	$d_{\mathcal{I}}$	r_{bud}	r_{burst}
(1, 0, 0.1)	1.10	0.417	0.250	0.180
(1, 0.5, 1/3)	1.00	0.910	0.395	0.294
(1, 0.9, 0.1)	0.42	0.701	0.343	0.198

Table 1.6: Deterministic growth rates at matched $R_0 = 2$, matched total yield and matched mean productive lifetime, with $c = 1$. Budding invades faster on both sides of the flooding boundary and on it.

The mechanism is generation time in the exact sense of section 1.4.2: the budding cell starts releasing at once while the bursting cell must first accumulate load, so budding's generation-time density $\mathcal{A}/R_0$ sits earlier, and by $\mathcal{A}(r) = 1$ an earlier density gives a larger root. This is the same delay that appeared as the eclipse-like lag of figures 1.6 and 1.7, and it is the phage lysis-timing problem in another notation: holding on longer makes more particles per cell but takes longer

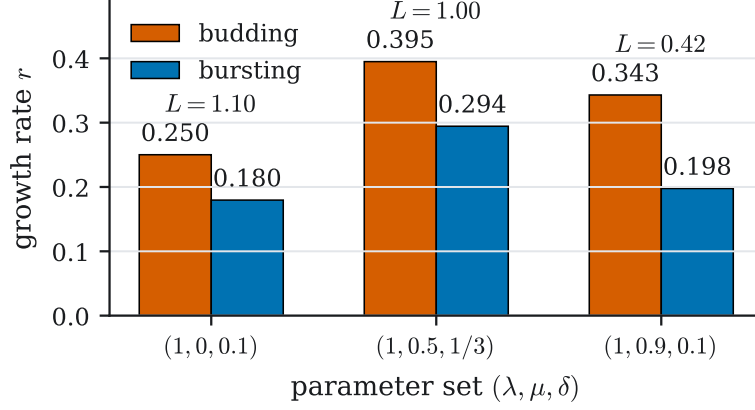


Figure 1.11: The same comparison as table 1.6, drawn. Budding invades faster than bursting in all three regimes of table 1.2, which have $L = 1.10, 1.00$ and 0.42 : the ordering does not turn at the flooding boundary, so it is not the same effect as theorem 1.8.

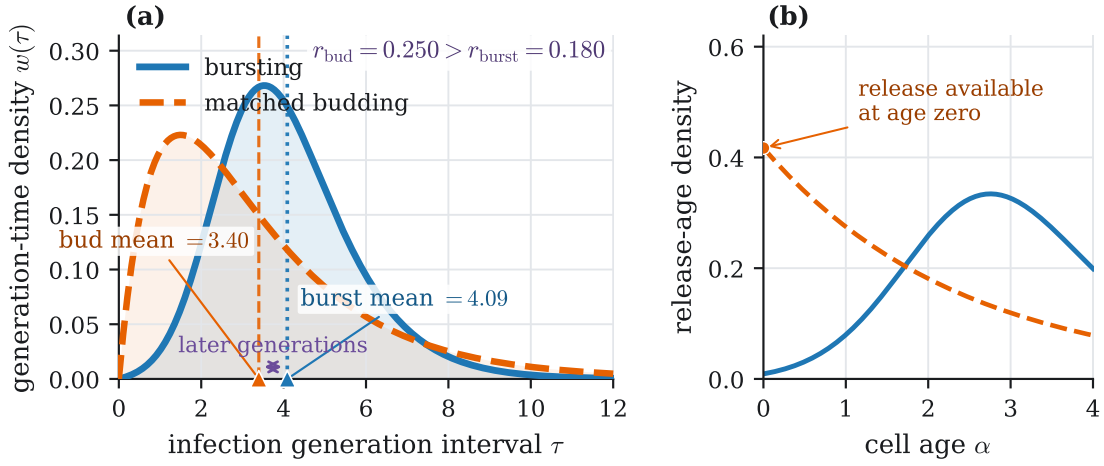


Figure 1.12: The generation-time structure behind the first row of table 1.6. At $(\lambda, \mu, \delta) = (1, 0, 0.1)$, budding and bursting are matched at $R_0 = 2$, $c = 1$, total yield and mean productive lifetime. The main curves are the normalised infection-generation laws $\mathcal{A}(\alpha)/R_0$, including the common extracellular waiting-time component; panel (b) isolates release age and shows why the budding law leads. Bursting’s mass sits in later generations: mean generation time 3.398 against 4.089, which is what puts $r_{\text{bud}} = 0.250$ above $r_{\text{burst}} = 0.180$.

to make them [12, 9]. That an infectious-period distribution shifts the growth rate at fixed threshold is a general fact about renewal epidemics rather than anything peculiar to bursting [13, 14]; what is particular here is that the distribution is derived rather than assumed.

Put together, the two comparisons say what bursting buys and what it costs. At fixed total output per cell, bursting can lower the probability of early extinction — when $L > 1$ — while slowing deterministic invasion: R_0 stays fixed, and z_{ext} and r move in opposite directions. Within this comparison, then, a pathogen tuning δ faces a genuine trade-off (figure 1.13). For $\mu > 0$ budding beats bursting on extinction until δ crosses $\delta_* = \lambda\mu/(\lambda + \mu)$, beyond which bursting takes over; for $\mu = 0$ bursting wins at every δ , at the continuing cost of a slower r .

Remark 1.11 (Scope of the comparison). Both results compare branching structures at fixed offspring mean. They say nothing about nonlinear effects, where other advantages of flooding may live; those are taken up in section 1.7.2. The evolutionary framing, and where this result sits relative to the bursting–budding spectrum, belongs to the chapter on bursting and budding

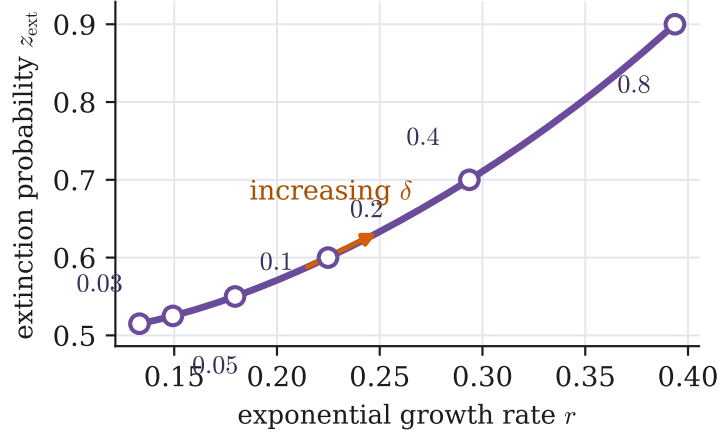


Figure 1.13: Tuning δ traces a strict extinction–growth trade-off, drawn parametrically as δ varies from 0.03 to 0.8 with $\lambda = 1$, $\mu = 0$, $c = 1$ and $R_0 = 2$. The coupling is reset along the curve to hold $R_0 = 2$, and the budding comparator is matched in lifetime yield V_∞ and in mean productive lifetime through $d_{\mathcal{I}} = 1/\mathbb{E}[T_{\text{prod}}]$ and $p = V_\infty d_{\mathcal{I}}$. There is no setting of δ that improves both coordinates: faster growth is bought with a higher extinction probability.

pathogenesis, which inherits from this comparison the awkward fact that one mechanism can be favoured on establishment and disfavoured on invasion speed at the same time — the nested-model problem in its usual form [15].

1.6 The reach of the construction

Section 1.3.3 showed that any pair of kernels induces the same population equations. Bursting and constant budding are the ends of a spectrum (figure 1.14); between them lie mechanisms that release many times per cell, release part of the load, or release in a self-excited cascade, and beyond it lie multi-stage models that are not birth–death–catastrophe processes at all. All supply a pair (S, g) , and none changes (1.13). The section closes on where that generality stops.

1.6.1 A finite-dimensional shadow: five ODE cases

When a fitting-ready ODE is wanted rather than a convolution, the same ideas assemble into a ladder of five cases (table 1.7), written out in section 1.E. Classical BMVR is Case 1 ((1.68)); stock-and-export recovers it in the fast-export limit ((1.72)); load-proportional dumps are Cases 4–5 ((1.74) and (1.77)). Each case also induces kernels, so (1.13) stays available whenever stage structure is traded for age structure.

1.6.2 Reset catastrophe: the worked intermediate

The absorbing process releases at most once per cell. The intermediate worth isolating lets a cell release *many* times: while the cell is alive the load undergoes birth λn , death μn , optional

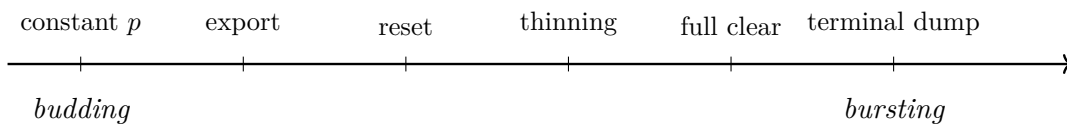


Figure 1.14: The budding–bursting spectrum. Every station is a choice of (S, g) .

Case	Intracellular story	Free-particle source
1 Classic	none (instant release)	$p\mathcal{I}$
2 Eclipse + classic	delay $E \rightarrow \mathcal{I}$, then as 1	$p\mathcal{I}$
3 Stock + continuous export	ν into Q , export εQ	εQ
4 Stock + full clear	ν into Q , dump $\propto Q^2/\mathcal{I}$	$\delta Q^2/\mathcal{I}$
5 Eclipse + stock + full clear	$E \rightarrow \mathcal{I}$, then as 4	$\delta Q^2/\mathcal{I}$

Table 1.7: The five ODE cases. Case 5 is Case 4 composed with Case 2’s eclipse block; the ladder is ordered by how load-dependent the release term is.

immigration ν , and a *dump that resets* — at rate δn the whole pool is released and the load returns to zero, after which it may re-accumulate — with the cell dying at a separate, load-independent rate $d_{\mathcal{I}}$. Reading the load as a local assembled pool rather than the whole-cell contents makes this a tractable multi-pulse caricature between full dumping and many small exports.

Because dumps do not absorb, the kernels change but the skeleton does not:

$$S(\alpha) = e^{-d_{\mathcal{I}}\alpha}, \quad g(\alpha) = \delta e^{-d_{\mathcal{I}}\alpha} \mathbb{E}[X_{\alpha}^2], \quad (1.41)$$

where $\mathbb{E}[X_{\alpha}^2]$ is computed for the non-absorbing reset process. Linear rates keep its moments tractable, so g is explicit once they are solved, and structurally $g \propto \mathbb{E}[X^2]$ exactly as for the absorbing process. The renewal system is identical in form to (1.10)–(1.11), with

$$p_{\text{eff}}(r) = (r + d_{\mathcal{I}}) \tilde{g}(r), \quad d_{\mathcal{I},\text{eff}} \equiv d_{\mathcal{I}}, \quad (1.42)$$

and $R_0 = (\gamma T/c) \int_0^{\infty} g$, finite only for $d_{\mathcal{I}} > 0$, since otherwise recurrent dumps could continue forever. If the reset process admits a stationary law π and $d_{\mathcal{I}}$ is small, the mature-cell limit is

$$p_{\infty} = \lim_{\alpha \rightarrow \infty} \frac{g(\alpha)}{S(\alpha)} = \delta \mathbb{E}_{\pi}[X^2] : \quad (1.43)$$

a long-lived infected cell behaves as classical constant- p BMVR with that p , while transients — first dumps, young cells — still need the full kernel. An eclipse is added as before: set $g(\alpha) = 0$ for $\alpha < \tau_e$ and keep S . This is what makes the spectrum claim concrete rather than gestural (table 1.8): an intermediate with kernels, an effective release rate, a mature limit, and a definite answer as to which classical model it converges on.

	Absorbing BDC	Reset catastrophe
After dump	process ends in H	$X \rightarrow 0$, continues
Dumps per cell	≤ 1	many, until cell death
Survival kernel	I_{fix}	$e^{-d_{\mathcal{I}}\alpha}$
Release kernel	$\delta K_{\text{BDC}}(\alpha)$	$\delta e^{-d_{\mathcal{I}}\alpha} \mathbb{E}[X_{\alpha}^2]$
Population equations	renewal BMVR	same form
Mature limit	not constant budding	$p \rightarrow \delta \mathbb{E}_{\pi}[X^2]$

Table 1.8: Absorbing against resetting catastrophe. Only the kernels differ; the population layer does not know which it has been given.

1.6.3 Partial release

The last axis is *how much* leaves the cell when a release event fires. Cases 3–5 sit at the extremes, continuous export and full clear; the natural intermediate removes a random part of the load.

Two ingredients are free: the event clock $\lambda(X)$ — constant δ , linear δX , or cooperative δX^2 — and the law of the amount released. The default is boolean thinning, each unit released independently with probability φ ; the other options are catalogued in [section 1.E](#). One result organises the axis. At the mean-field level with $\lambda = \delta X$, the population equations are Case 4 scaled by φ :

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \varphi \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \varphi \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (1.44)$$

With a constant clock $\lambda = \delta$ instead, the mean export is $\delta \varphi Q$ and the bulk equations collapse to Case 3 with $\varepsilon_{\text{eff}} = \delta \varphi$. So partial release is invisible in the mean-field equations whenever the event clock is load-independent, and visible when it is load-dependent; in single-cell release-count distributions it is always visible. Where in φ the flooding advantage of [theorem 1.8](#) survives is open, and the closed-form structure may not persist at intermediate φ at all.

1.6.4 Self-exciting release

Release may also be partially *self-exciting*: one event makes further release more likely for a short time, through cooperative nucleation, local membrane platforms, or machinery already recruited at a site. Open-ended self-excitation is implausible, since stock depletes and machinery is finite, but the skeleton accommodates any hybrid of excitation, refill and depletion, because a renewal model exists whenever $g(\alpha) = \mathbb{E}[\lambda(\alpha) \mid \text{alive}]$ is computable. A finite ODE model needs the excitation carried by a low-dimensional Markov state. The exponential-kernel Hawkes process [\[16\]](#) is the case in which it is: Markovian in a shot-noise intensity, and so ODE-representable. The general event-history form is not. [Section 1.E](#) catalogues twelve candidates and develops four.

1.6.5 A non-BDC instance: HIV

This chapter’s results are written for pathogens whose intracellular life ends in load-proportional catastrophe. HIV is not one: it buds from the plasma membrane, and an infected cell need not rupture in one event. Yet single-cell and limiting-dilution data show something surge-like — long quiet periods, then a concentrated bout of release — which is the phenomenon the birth–death–catastrophe process exists to explain. The position taken here is neither to force HIV into that process nor to retreat to constant- p BMVR, but to build an eclipse-structured, short-productive-pulse model on the same skeleton.

Hataye et al. [\[17\]](#) coupled limiting-dilution latency-reactivation cultures to stochastic models. The cultures show delay plus a short pulse, not a terminal dump: first detection around day 3, bulk production over days 3–5, an Erlang eclipse with $n \approx 5$, and many silent lineages ($f_{\text{HIV}} \approx 0.41$). Establishment against the number of virus-releasing cells is sigmoid, which they read as an Allee effect, with threshold $\Lambda \approx 2.3$ cells (about 5100 RNA copies under a $\geq 2 \times 10^5$ definition) and about 2% establishment from one stimulated cell. Those numbers are collected in [table 1.13](#); [tables 1.9](#) and [1.10](#) record what they support and what they do not.

Candidate stage structure. What the data ask for is a two-path eclipse feeding a short productive class, drawn in [figure 1.15](#): a stimulated latent cell L_0 enters either a multi-stage path A of high release potential or a brief path B of low potential, eclipse cells divide or die without releasing, and productive cells release at constant rate until they die. The reaction list and the stage equations it induces are [section 1.F](#); only two features matter here. The delay comes from the length of the chain rather than from any slow release rate, and eclipse proliferation is a genuine source term rather than a relabelling — which is what lets a single reactivated lineage produce multi-day detection without any one cell budding for days.

Supported for HIV	Not supported as literal HIV
Multi-stage eclipse (delay)	Pure one-type BDC terminal catastrophe
Short productive pulse then cell death	Memoryless constant- p BMVR from $t = 0$
High stochasticity; many silent lineages	Independent branching alone near threshold
Renewal / stage structure for free virus	QSD = geometric burst size as an HIV law
Allee / synergistic establishment	

Table 1.9: What the limiting-dilution data support, and what they do not. The right-hand column is the list of claims this chapter does not make for HIV.

	Lytic BDC (this chapter)	HIV (Hataye-guided)
Release mechanism	catastrophe at rate δX	budding while productive
End of production	terminal; process absorbed	short productive life; cell dies
Intracellular load X	explicit birth–death state	not required as primary state
Quiet period	until first catastrophe	multi-stage eclipse (+ latency)
Sustained multi-day release	one large dump (or nothing)	often staggered short pulses
Establishment theory	independent offspring as baseline	Allee synergy near threshold
Natural hosts	<i>Y. pestis</i> , <i>F. tularensis</i> , phage	CD4 T-cell latency / rebound

Table 1.10: Two mechanisms, compared where they differ. The seam is the release event: terminal and load-proportional on the left, repeated and load-independent on the right.

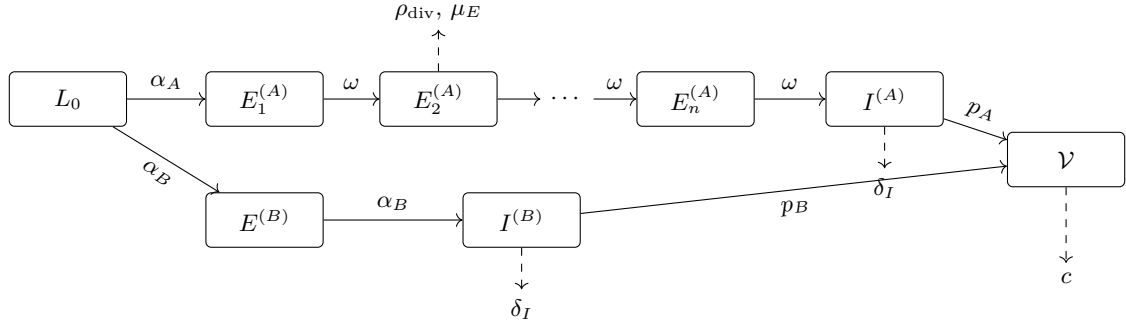


Figure 1.15: The stage structure of (1.80): two eclipse pathways from the stimulated latent class L_0 — a multi-stage path A of high release potential and a brief path B — each feeding a short productive class releasing at constant rate. Division and death (dashed) act in eclipse, death δ_I in the productive classes, clearance c on free virus. The source of the delay is the chain, not the release rate.

For ongoing infection the initial cohort becomes incidence, split between pathways with $\pi_A + \pi_B = 1$:

$$i(t) = f(\mathcal{V}(t), T), \quad \text{source into } E_1^{(A)} = \pi_A i(t), \quad \text{source into } E^{(B)} = \pi_B i(t). \quad (1.45)$$

Linear infection, $f(\mathcal{V}, T) = \gamma T \mathcal{V}$, recovers a stage-structured BMVR. Near the threshold, though, the data favour a sigmoid law over independent per-virion success; the minimal upgrade is the Allee incidence

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \quad (1.46)$$

inefficient at very low $\mathcal{V}$ and saturating to mass action for $\mathcal{V} \gg K_A$, with K_A set so that cumulative initial release near $\sim 5 \times 10^3$ RNA copies sits at the inflection of establishment probability. More mechanistic forms can replace it; the point is that linearity in $\mathcal{V}$ alone is not faithful near establishment. Writing g_{HIV} for the expected release rate at age α after a single infection or reactivation and S_{HIV} for the probability of still being in a productive lineage, (1.13)

holds exactly:

$$\mathcal{I}(t) = \mathcal{I}_0 S_{\text{HIV}}(t) + (i * S_{\text{HIV}})(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g_{\text{HIV}}(t) + (i * g_{\text{HIV}})(t) - c \mathcal{V}(t). \quad (1.47)$$

Here g_{HIV} is *not* δK : it is the density of productive sojourn times, weighted by p_A and p_B , with the Erlang eclipse supplying the delay — near zero for small α , then rising and falling, which is the mathematical expression of nothing for a long time, then a lot. The effective parameters carry over under $(I_{\text{fix}}, g) \mapsto (S_{\text{HIV}}, g_{\text{HIV}})$:

$$p_{\text{eff}}(r) = \frac{\tilde{g}_{\text{HIV}}(r)}{\tilde{S}_{\text{HIV}}(r)}, \quad R_0 = \frac{\gamma T}{c} \int_0^\infty g_{\text{HIV}}(\alpha) d\alpha \quad (\text{linear incidence}), \quad (1.48)$$

while under (1.46) R_0 remains the large-population threshold and early establishment is governed in addition by the scale K_A .

Linear or exponential intracellular growth. HIV does not amplify infectious units by branching: after integration the template count is one or a few proviruses, and structural products do not self-replicate. Production is immigration-like: the mean load follows $\bar{x}' \approx \nu - \mu \bar{x}$, set by a fixed template count, rather than the $\bar{x}' = r \bar{x}$ of a process in which each unit begets more units. That is the whole reason the release kernel cannot be δK . The regime adopted here is eclipse then mean-linear accumulation.

1.6.6 Where the reach stops

The skeleton is general; the results proved on it are not. What carries over to any single-cell mechanism is (1.13), the map $p_{\text{eff}} = \tilde{g}/\tilde{S}$, the reading of constant p as an exponential-phase projection, and the generation-kernel formulation of R_0 and r . What does not is everything resting on the closed forms of the birth–death–catastrophe process: the flooding criterion $1/\delta < 1/\lambda + 1/\mu$, the geometric burst law and its identification with the quasi-stationary distribution, one-shot cumulative release as the only single-cell trajectory. For HIV, independent branching is not the full establishment story near the observed threshold; the flooding comparison survives as a *linear* baseline, at matched mean yield with offspring variances compared. But near $\sim 5 \times 10^3$ RNA copies synergy can dominate variance effects, and the two mechanisms must be separated before either is invoked.

1.7 Discussion

This chapter took the completed single-cell theory of the chapter on the birth–death–catastrophe process and the chapter on its distribution theory and embedded it in a population. What came out is a renewal model of within-host infection, forced by the intracellular process and exact in its kernels; the demonstration that classical BMVR is a projection of it, valid one exponential phase at a time; and an effective-parameter map whose two ends differ by orders of magnitude. Three consequences came with them: R_0 is invariant under the replacement of budding by bursting, the flooding advantage at establishment is paid for in invasion speed, and the construction is general enough to hold every release mechanism of section 1.6 in one frame.

1.7.1 Consequences for fitting practice

Three results of section 1.4 bear on how within-host models are fitted, and each says what to do rather than only what is true.

A fitted p is a property of the epidemic’s phase, not of the pathogen, so acute-phase and set-point estimates should differ systematically, in a direction and by an amount (1.15) specifies.

Two published estimates of p for the same organism at different phases are therefore not in conflict, and the ratio between them is itself an observable: by (1.22) it cannot exceed $\mathbb{E}[X^2]_{\text{QS}}$, so a reported spread larger than that is evidence against the intracellular model rather than against either fit.

The eclipse-like delay that within-host models insert by hand emerges here from the intracellular dynamics with no extra parameter, its duration set by the mean productive lifetime of (1.57). A model already carrying a fitted eclipse and a bursting pathogen is thus fitting the same delay twice, and the two estimates should be checked against each other before either is trusted.

And no fit of (p, d_T) recovers the three intracellular rates (section 1.4.5). The practical consequence is that a population fit is worth doing only alongside one of the three single-cell observables named there; with any one of them the system is over-determined, and the fit becomes a test of the intracellular model instead of a description of it. Which of the three is easiest to measure is weighed in the chapter on its distribution theory.

1.7.2 Limitations

The assumptions of section 1.1.2 are load-bearing, and each would change something definite if relaxed. Holding target cells constant (A1) makes the model linear, so supercritical solutions grow without bound and the peak times of section 1.4.3 exist only in the subcritical cases; depletion would cap them, but the linear analysis remains the right object for establishment and early growth, where the chapter's claims live. The continuum convention (A2) is why $R_0 = \gamma TV_\infty / c$ sits alongside the branching mean $m = qV_\infty$: they agree when $\gamma T \ll c$ and are otherwise different quantities, and confusing them would misread theorem 1.8. The synchronous initial condition (A3) is not innocuous, since a non-trivial initial age distribution adds a term to (1.10). Single infection (A4) is a genuine restriction — multiply infected cells release superlinearly more, by (1.63) and (1.64) — and the multiplicity kernels exist but have not been carried through the population layer. Absence of spatial structure (A5) is the most consequential omission for a pathogen that spreads cell to cell.

Two further boundaries are worth naming. The branching comparison of section 1.5 holds the offspring mean fixed and is silent about nonlinear effects; and the verification of section 1.C covers reinfection only indirectly, through the characteristic equation.

The chapter is also written in dimensionless rates with $\lambda = 1$, and does not say where a real pathogen sits on figure 1.9. For *Y. pestis* in the macrophage the interval is short and the biology resists a clean measurement: the bacterium enters a permissive vacuole and replicates there while the shift from the flea's ambient 26–28°C to the host's 37°C induces the capsule, the pH 6 antigen and the type III secretion system, and then leaves by a route the primary literature still treats as unresolved, with apoptosis, pyroptosis and necroptosis all reported. Placing that organism on the $L = 1$ boundary needs λ , μ and δ for the intracellular phase — and the exit mechanism is exactly what fixes δ .

1.7.3 Open problems

Four questions remain open on the population side.

1. **Two-type killed second moments, and the two-type D .** The release kernel of a two-type renewal model needs $\mathbb{E}[X_\alpha^2 \mathbf{1}\{\tau_c > \alpha\}]$, $\mathbb{E}[X_\alpha Y_\alpha \mathbf{1}\{\tau_c > \alpha\}]$, $\mathbb{E}[Y_\alpha^2 \mathbf{1}\{\tau_c > \alpha\}]$, and the analogue of $D(t)$, extinction without catastrophe. The moment hierarchy does not close upward, exactly as in one type; the one-type escape was to differentiate the Riccati equation, and the two-type route should be to differentiate the backward system of the two-type chapter. This is the main technical obstacle between this chapter and an eclipse-aware renewal model, and it should not be assumed easy.

2. **The flooding boundary against real parameters.** Map $\delta(\lambda + \mu) > \lambda\mu$ for *Y. pestis* in the macrophage, and compute the *size* of the advantage from (1.35): does the pathogen sit on the advantageous side, and by how much? The estimates discussed in the chapter on its distribution theory — the near-certain bursting of *F. tularensis* [18], the clearance-dominated anthrax case — suggest that real pathogens sit on both sides of the boundary, which is what makes the question worth asking.
3. **Population-level variance.** Propagate $\text{Var}(W)$ of (1.61) through the branching structure to a whole-host prediction. For single-cell assays the first moments suffice; for whole-host data the variance is the main stochastic signal that classical BMVR cannot reproduce. The integer-valued nature of the offspring count, whose distribution matters for establishment [19], enters through the same quantity.
4. **Partial release and the flooding boundary in φ .** Where on the continuum of section 1.6.3 does the flooding advantage survive, and does the closed-form structure persist at intermediate φ ?

Two smaller questions are recorded in the text where they arise, because each is now a single definite statement rather than an open direction. The hypothesis of proposition 1.6 — that $\rho = g/S$ increases with cell age — is a claim about the one closed-form function (1.21) of one variable, so proving it is finite algebra; and the ordering $r_{\text{bud}} > r_{\text{burst}}$ is by proposition 1.10 equivalent to the single Laplace inequality (1.40), with the coupling constants eliminated. Both are supported by wide numerical sweeps (tests M and N of section 1.C) and neither is proved. The single-cell open problems belong to the chapter on its distribution theory.

1.7.4 Connections forward

The two-type process of the two-type chapter supplies the most immediate continuation, and it is where the chapter’s principal technical obstacle lives. In the GATE regime $\delta_1 = 0 < \delta_2$ — no burst before conversion — the no-catastrophe probability has zero initial hazard and then rises: precisely the eclipse-phase signature that within-host models insert by hand, emerging mechanistically, with the eclipse duration set by the conversion rate. The construction here is ready to receive the two-type kernels the moment they exist; nothing in (1.13) need change to accept them. The bursting–budding comparison finds its evolutionary home in the chapter on bursting and budding pathogenesis, where the intermediate mechanisms of section 1.6 supply the phenotypes a selection argument needs. The nonlinear effects that the later modelling chapters treat — immune saturation, local depletion, the all-run-out-at-once intuition — are exactly what the branching comparison deliberately excludes, and should be named wherever the flooding advantage is invoked.

1.A Results quoted from earlier chapters

The body uses the single-cell closed forms of the chapter on the birth–death–catastrophe process constantly and the distribution-level facts of the chapter on its distribution theory at isolated points. This appendix quotes both, so that the introduction need not reprint them. Figure 1.16 records what is handed across.

1.A.1 Roots, identities and moments

Write $w = e^{\theta t}$. The roots of the quadratic $\lambda f^2 - (\lambda + \mu + \delta)f + \mu$ are

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda}, \quad (1.49)$$

with $b < 1 < a$, and the derived constants are

$$A = a - 1, \quad B = 1 - b, \quad \theta = \lambda(a - b), \quad \kappa = 1 + \frac{\delta}{2\lambda}. \quad (1.50)$$

The identities that simplify everything downstream are

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}, \quad AB = \frac{\delta}{\lambda}, \quad A + B = a - b, \quad \theta = \lambda(a - b). \quad (1.51)$$

The no-burst probability, the internal-extinction probability, and the non-fixation probability are

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad D(t) = \frac{ab(w - 1)}{aw - b}, \quad I_{\text{fix}}(t) = I(t) - D(t) = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}. \quad (1.52)$$

Here $I(\infty) = D(\infty) = b$, the probability that the cell clears internally without ever bursting, while $I_{\text{fix}}(0) = 1$, $I'_{\text{fix}}(0) = -(\mu + \delta)$ and $I_{\text{fix}}(\infty) = 0$. With $I' = -\delta J$ and $V' = \delta K$,

$$J(t) = \frac{(a - b)^2 w}{(B + Aw)^2} = -\frac{\lambda}{\delta}(I - a)(I - b), \quad (1.53)$$

$$K(t) = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right]J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad (1.54)$$

and the cumulative mean release is a quadratic in $1 - I$ alone,

$$V(t) = (1 - I)\left[1 + \frac{\lambda}{\delta}(1 - I)\right]. \quad (1.55)$$

Its late-time value, the total expected output of one cell, is

$$V_{\infty} = \frac{a(1 - b)}{a - 1} = \frac{aB}{A} = \frac{\lambda - \mu}{\delta}(1 - b) + 1, \quad \mathbb{E}[\mathcal{K} \mid \text{burst}] = \frac{V_{\infty}}{1 - b} = \frac{a}{a - 1}, \quad (1.56)$$

in which V_{∞} counts the internally cleared cells with their release of zero and the conditional mean removes them. The integral of the survival kernel, which anchors [section 1.4](#), is

$$\mathbb{E}[T_{\text{prod}}] = \int_0^{\infty} I_{\text{fix}}(\alpha) d\alpha = \frac{1}{\lambda} \log \frac{a}{a - 1} = \frac{1}{\lambda} \log \mathbb{E}[\mathcal{K} \mid \text{burst}]. \quad (1.57)$$

1.A.2 Distribution-level facts

Three results of the chapter on its distribution theory are used as black boxes. First, the burst-time density is

$$-I'(t) = \delta J(t), \quad (1.58)$$

defective, with total mass $1 - b$. Second, the burst size, unconditional, obeys

$$\mathbb{P}\{\mathcal{K} = k\} = \frac{\delta}{\lambda} a^{-k}, \quad k \geq 1, \quad (1.59)$$

with the mass b at “no burst”; conditioned on bursting this becomes $(a - 1)a^{-k}$, which is exactly the quasi-stationary distribution of the load [\[20\]](#) — both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$. Third, given that the burst occurs at time t , its size is a size-biased draw from the state probabilities:

$$\mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{K(t)}{J(t)} = 1 + \frac{2\lambda}{\delta}(1 - I(t)), \quad (1.60)$$

tending at late time to $(a + 1)/(a - 1)$, the size-biased quasi-stationary mean.

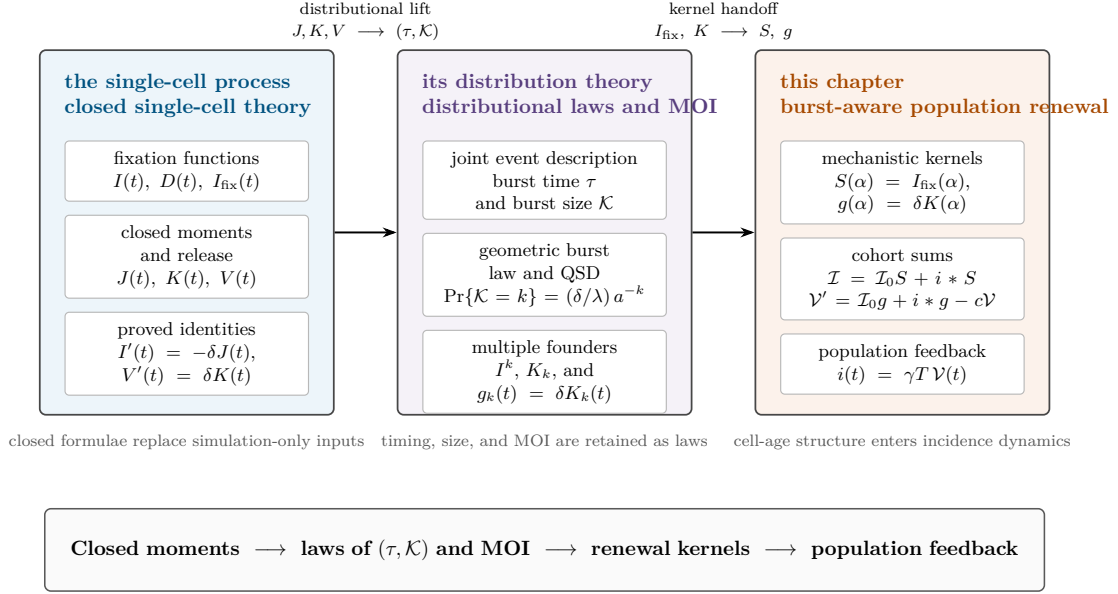


Figure 1.16: What each chapter hands to the next: closed single-cell moments and fixation functions from the chapter on the birth–death–catastrophe process, distributional laws and multiplicity from the chapter on its distribution theory, and burst-aware renewal kernels in population dynamics here.

1.A.3 The second moment of the release

The whole-host variance prediction of [section 1.7.3](#) starts from

$$\mathbb{E}[W_t^2] = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1, \quad (1.61)$$

with $\text{Var}(W_t) = \mathbb{E}[W_t^2] - V^2$. Its limit supplies $\text{Var}(\mathcal{K})$, and hence the offspring variance (1.32).

1.A.4 Multiplicity of infection

A cell founded by k particles releases superlinearly more. Its moments and release flux are

$$J_k(t) = kJ(t)I(t)^{k-1}, \quad K_k(t) = kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2}, \quad (1.62)$$

$$g_k(t) = \delta K_k(t) = \delta \left(kK I^{k-1} + k(k-1)J^2 I^{k-2} \right), \quad (1.63)$$

and its lifetime yield, for $k \geq 2$,

$$V_\infty^{(k)} = (1 - b^k) + \frac{2\lambda}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\lambda}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx, \quad (1.64)$$

simplifying to $V_\infty^{(k)} = k + \lambda/\delta$ when $\mu = 0$. The conditional mean burst $V_\infty^{(k)}/(1 - b^k)$ increases with k : multiply infected cells burst more often, and bigger. The pair $(I(\alpha)^k - D(\alpha)^k, \delta K_k(\alpha))$ is the MOI- k row of [table 1.4](#); assumption (A4) of [section 1.1.2](#) sets it aside for the population layer.

1.B Master formula table: population level

Throughout: $w = e^{\theta t}$, $\theta = \lambda(a - b)$, $A = a - 1$, $B = 1 - b$, $L = a(1 - b)$, $\kappa = 1 + \delta/(2\lambda)$. The single-cell rows are collected in the chapter on its distribution theory’s formula appendix; [table 1.11](#) records the population-level quantities of the present chapter.

Quantity	Formula
Renewal system	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}} + i * I_{\text{fix}}, \mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}, i = \gamma T\mathcal{V}$
Survival kernel	$S(\alpha) = I_{\text{fix}}(\alpha)$
Release kernel	$g(\alpha) = \delta K(\alpha) = V'(\alpha)$
Generation kernel	$\mathcal{A}(\alpha) = \gamma T(g * e^{-c \cdot})(\alpha), \tilde{\mathcal{A}}(r) = \gamma T\tilde{g}(r)/(r + c)$
$p_{\text{eff}}(r)$	$\delta \tilde{K}(r)/\tilde{I}_{\text{fix}}(r)$
$d_{\mathcal{I}, \text{eff}}(r)$	$1/\tilde{I}_{\text{fix}}(r) - r$
Limits $r = 0$	$p_{\text{eff}} = \lambda V_{\infty}/\log(a/(a-1)), d_{\text{eff}} = \lambda/\log(a/(a-1))$
Limits $r \rightarrow \infty$	$p_{\text{eff}} \rightarrow \delta, d_{\text{eff}} \rightarrow \mu + \delta$
Characteristic eq.	$r + c = \gamma T \delta \tilde{K}(r), \text{ equivalently } \tilde{\mathcal{A}}(r) = 1$
R_0^{ODE}	$\gamma T V_{\infty}/c = \int_0^{\infty} \mathcal{A}$
Branching mean	$m = qV_{\infty}, q = \gamma T/(\gamma T + c)$
$G_{\text{off}}(z)$	$b + (\delta/\lambda) y/(a - y), y = 1 - q + qz$
$z_{\text{ext}}^{\text{burst}}$	$[a(1 - q + qb) - 1 + q]/q$
$z_{\text{ext}}^{\text{bud}}$	$1/m$ (geometric output, matched mean)
Flooding criterion	$1/\delta < 1/\lambda + 1/\mu \ (L > 1)$
$L = 1$	offspring law exactly geometric
Δz_{ext}	$(L - 1)(1/m - 1)$
$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}}$	$2q^2 V_{\infty}(L - 1)/(a - 1)$
Growth trade-off	$r_{\text{bud}} > r_{\text{burst}}$ at matched $R_0, \mathbb{E}[T_{\text{prod}}]$
HIV stage model	kernels $(S_{\text{HIV}}, g_{\text{HIV}})$; $p_{\text{eff}} = \tilde{g}_{\text{HIV}}/\tilde{S}_{\text{HIV}}$; Allee incidence $f = \gamma T\mathcal{V}^2/(K_A + \mathcal{V})$

Table 1.11: Population-level quantities of this chapter, in one place.

Quick-reference specialisations

$\mu = 0$. $b = 0, a = 1 + \delta/\lambda, D \equiv 0, I_{\text{fix}} = I, V_{\infty} = 1 + \lambda/\delta = \langle X \rangle_{\text{QS}}, \mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log(1 + \lambda/\delta), \mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$ with $s = \delta/(\lambda + \delta), \rho = 1 - s$; flooding favours bursting for establishment at every $\delta > 0$.

The boundary $L = 1$. $\delta = \lambda\mu/(\lambda + \mu), b = \delta/\lambda$; the offspring law is geometric with mean m ; $z_{\text{ext}}^{\text{burst}} = z_{\text{ext}}^{\text{bud}}$ for every q ; offspring variances equal.

Young-cell, fast-growth limit. $p_{\text{eff}} \rightarrow \delta, d_{\text{eff}} \rightarrow \mu + \delta$: a newly infected cell holds one particle, bursts at rate δ , releases one.

Set-point, $r = 0$ matching. $p_{\text{eff}} = V_{\infty}/\mathbb{E}[T_{\text{prod}}], d_{\text{eff}} = 1/\mathbb{E}[T_{\text{prod}}]$: lifetime output over lifetime; the fairest constant-parameter comparison, and the one used in [section 1.4.3](#).

1.C Verification record

Every quantitative claim of this chapter was checked analytically where possible and numerically where not. The suite tests (1.10)–(1.11) at four levels: the integrity of the kernels; the reduction to single-cell formulae without reinfection; the structural limits available in closed form; and an independent stochastic simulation of a finite cohort. The parameter sets are $(\lambda, \mu, \delta) \in \{(1, 0, 0.1), (1, 0.2, 0.05), (1, 0, 0.5), (1, 0.9, 0.1), (1, 0.5, 1/3)\}$ — the three regimes of [table 1.2](#) and two further sets chosen to spread the kernel shapes. Fifty-four checks, all passing ([table 1.12](#)).

Three checks are shown rather than tallied, because what they establish is used in the body: [figure 1.17](#) is test D, that the construction contains the classical model exactly rather than

ID	Name	What is checked	Result
A	Kernel self-consistency	$V' = g, \int g = V_\infty, \int I_{\text{fix}} = \mathbb{E}[T_{\text{prod}}], I' = -\delta J$	5/5
B	$\gamma T = 0$ cohort	$\mathcal{I} = \mathcal{I}_0 I_{\text{fix}}; \mathcal{V}' = \mathcal{I}_0 g - c\mathcal{V}$ vs exact convolution; $c = 0 \Rightarrow \mathcal{V} = \mathcal{I}_0 V(t)$	15/15
C	V-equation residual	$\mathcal{V}' - \mathcal{I}_0 g - \gamma T(\mathcal{V} * g) + c\mathcal{V} \approx 0$ on supercritical / subcritical / virion-seed solutions	5/5
D	Exponential kernels $\equiv$ classical	$S = e^{-d\alpha}, g = pe^{-d\alpha} \Rightarrow$ renewal = BMVR ODE (rel. err. $< 5 \times 10^{-5}$)	3/3
E	Growth = characteristic root	late $\log \mathcal{V}$ slope equals r_* solving $r + c =$ $\gamma T \delta \tilde{K}(r)$ (rel. err. $\sim 10^{-6}$ – 10^{-7})	5/5
F	R_0 threshold	$R_0 \in \{0.5, \dots, 3\}$: decay iff $R_0 < 1$, growth iff $R_0 > 1$	8/8
G	$p_{\text{eff}}(r_*)$ match	classical ODE with $(p, d) = (p_{\text{eff}}(r_*), d_{\text{eff}}(r_*))$ shares r_* with the renewal system	4/4
H	Gillespie cohort	independent BDC cells + clearance; ensemble mean vs renewal (rel. $L^2 \sim 1$ – 2%)	2/2
I	Virion-only seed	early $\mathcal{V} \approx V_0 e^{-ct}; \mathcal{I}(0) = 0$ then rises	1/1
J	Lifetime output	$\gamma T = c = 0: \mathcal{V}(\infty) = V_\infty$ (rel. err. $< 10^{-6}$)	5/5
K	Well-posedness	non-negative, finite trajectories	1/1
L	Old-cell limit	$g/S \rightarrow \delta \mathbb{E}[X^2]_{\text{QS}} = \delta a(a+1)/(a-1)^2$ at large age (rel. err. $< 10^{-8}$)	5/5
M	Monotonicity	$\rho = g/S$ non-decreasing in age over 4000 ran- dom (μ, δ) ; p_{eff} strictly decreasing on an r grid	2/2
N	Laplace order	$\tilde{g}_{\text{burst}}(r) \leq \tilde{g}_{\text{bud}}(r)$ at 60 values of r over 1500 random (μ, δ)	1/1
Total			62/62

Table 1.12: The verification catalogue. Tests A–C check the inputs and the bookkeeping; D–G the structural limits; H–K the behaviour of solutions; L–N the three results of [propositions 1.5, 1.6](#) and [1.10](#).

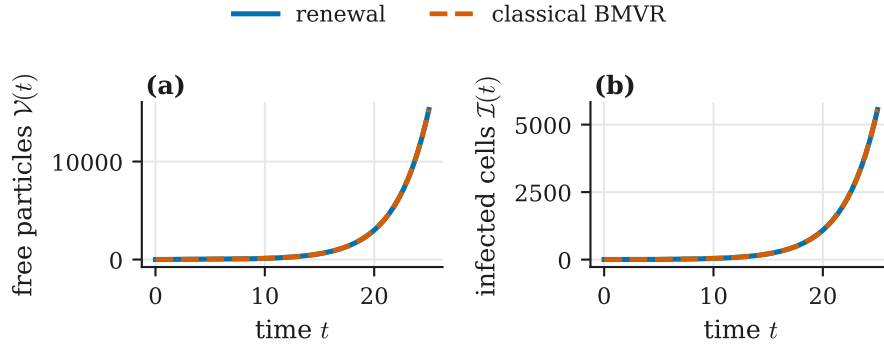


Figure 1.17: Test D. When the kernels are pure exponentials, $S = e^{-d\alpha}$ and $g = pe^{-d\alpha}$, the renewal system *is* classical BMVR with those constants — the two curves are the same solution, to relative discrepancy below 5×10^{-5} , for three matched parameter pairs, in (a) the free particles and (b) the infected cells. This is what makes [section 1.4](#) a projection rather than an approximation.

approximately, and [figure 1.18](#) is tests E and F, the growth rate and the threshold.

The reinfection case ($\gamma T > 0$) is checked indirectly, through the characteristic equation — tests C, E, F and G — which any correct linear theory must satisfy; a direct multi-generation stochastic check is left to future work.

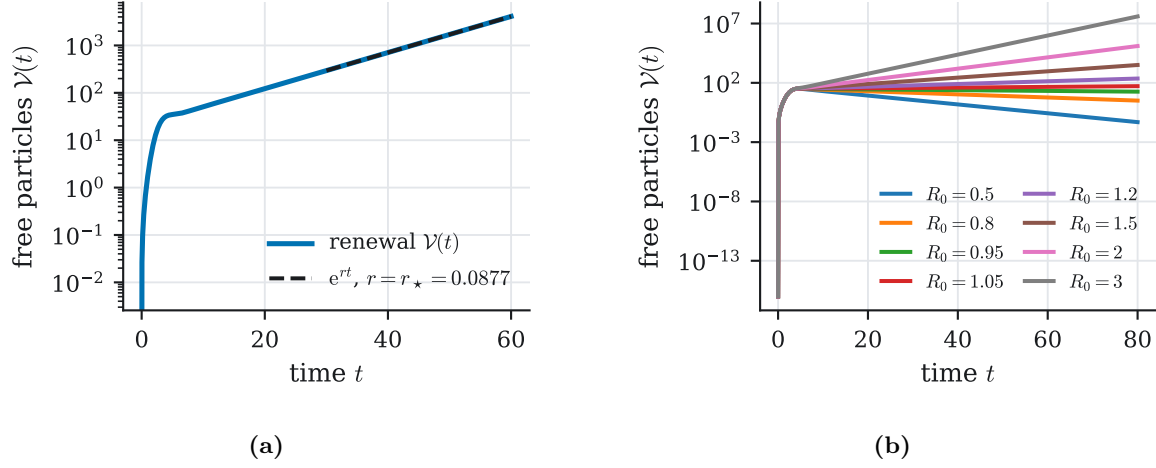


Figure 1.18: Tests E and F. (a) The fitted late growth rate of $\mathcal{V}$ against the root of (1.16), relative error $\sim 10^{-6}$ – 10^{-7} across five scenarios — the numerical content of (1.24). (b) The R_0 threshold scan, eight values through $R_0 = 1$, with the transition resolved already at $R_0 = 1.05$; $(\lambda, \mu, \delta) = (1, 0, 0.1)$.

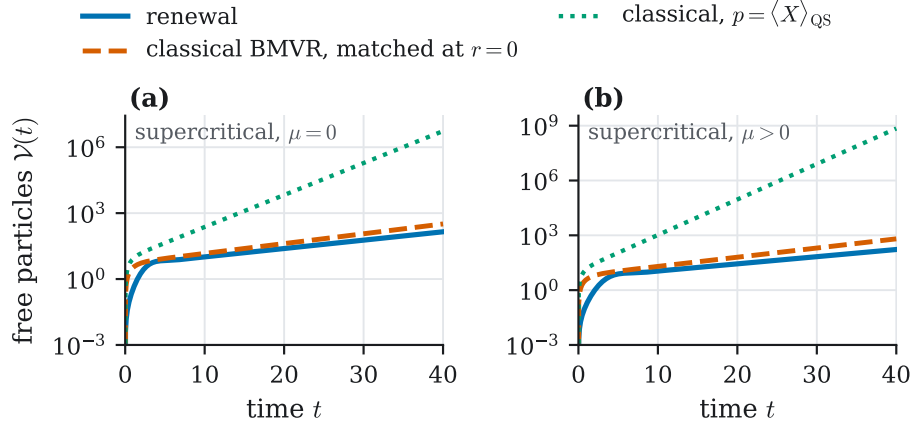


Figure 1.19: What a dimensional error looks like on a plot. Free particles under the renewal system (solid), under classical BMVR matched at $r = 0$ (dashed), and under the proposal of (1.5) (dotted): classical BMVR run with $p = \langle X \rangle_{\text{QS}}$, a pure count used as a rate. It is wrong in scale, not merely in shape. (a) $(\lambda, \mu, \delta) = (1, 0, 0.1)$ with $\gamma T = 0.04$, $c = 0.25$; (b) $(1, 0.2, 0.05)$ with $\gamma T = 0.03$, $c = 0.2$.

Reproduction. The suite is `verification/verify_result_20_1.py` in the chapter source (Python 3.14, NumPy 2.4, Matplotlib 3.10). From the chapter directory:

```
python3 verification/verify_result_20_1.py
# writes verify_result_20_1_report.txt, metrics, verify_figures/*.pdf
```

Run under Python 3.14.6, NumPy 2.4.2 and Matplotlib 3.10.8; the suite has no other dependency.

1.D Technical derivations

1.D.1 Hypergeometric Laplace transforms of the kernels

Write $s := r/\theta$ and recall $\theta = \lambda(a-b)$. Both transforms reduce, under $v = e^{-\theta t}$ (so $dt = -dv/(\theta v)$, $v: 1 \rightarrow 0$), to moments of the form $\int_0^1 v^{s-1}(1-v)^n(A+Bv)^{-m}dv$, which are hypergeometric.

For the survival kernel, from (1.52),

$$I_{\text{fix}}(t) = \frac{(a-b)^2 w}{(B+Aw)(aw-b)}, \quad w = e^{\theta t},$$

and in $v = 1/w$,

$$\widetilde{I}_{\text{fix}}(r) = \frac{a-b}{\theta} \int_0^1 v^s \left[\frac{B}{A+Bv} + \frac{b}{a-bv} \right] dv,$$

using the partial-fraction decomposition $(a-b)/[(A+Bv)(a-bv)] = B/(A+Bv) + b/(a-bv)$ — both sides multiplied out give $(a-b)^2$ in the numerator after using $Ab+aB=a-b$. Each term is a standard hypergeometric integral, $\int_0^1 v^s/(1-zv)dv = (s+1)^{-1} {}_2F_1(1, s+1; s+2; z)$, giving

$$\widetilde{I}_{\text{fix}}(r) = \frac{1}{\lambda(s+1)} \left[\Psi(A, B, s) + \frac{b}{a} {}_2F_1\left(1, s+1; s+2; \frac{b}{a}\right) \right], \quad (1.65)$$

where, so that the hypergeometric series always converges,

$$\Psi(A, B, s) = \begin{cases} {}_2F_1(1, s+1; s+2; -\frac{A}{B}), & B \geq A, \\ \frac{B}{A} {}_2F_1(1, s+1; s+2; -\frac{B}{A}), & A > B, \end{cases} \quad (1.66)$$

the two branches being equal by the identity $(B/A){}_2F_1(1, s+1; s+2; -B/A) = {}_2F_1(1, s+1; s+2; -A/B)$ — both sides equal $(s+1)B \int_0^1 v^s/(A+Bv)dv$. (The single-argument form $(B/A){}_2F_1(1, s+1; s+2; -B/A)$, valid by analytic continuation for all $A, B > 0$, has a convergent defining series only when $B \leq A$, that is $\delta \geq \lambda - \mu$.) For the release kernel, $g = \delta K$ with $K = [1 + 2\lambda(1-I)/\delta]J$; the same substitution and termwise integration give

$$\delta \widetilde{K}(r) = \frac{\delta}{\lambda} \left[\frac{{}_2F_1(1, s; s+2; z)}{A(s+1)} + \frac{2 {}_2F_1(2, s; s+3; z)}{A^2(s+1)(s+2)} \right], \quad z = -\frac{B}{A}, \quad (1.67)$$

again with the understanding that for $B > A$ the defining series of the right-hand side must be evaluated through its analytic continuation, or through the integral representation from which it came. Both formulae reduce correctly at $s = 0$: with ${}_2F_1(1, 1; 2; z) = -\log(1-z)/z$, (1.65) gives $\widetilde{I}_{\text{fix}}(0) = \lambda^{-1} \log(a/(a-1)) = \mathbb{E}[T_{\text{prod}}]$, and (1.67) gives $\delta \widetilde{K}(0) = (\delta/\lambda)(1/A + 1/A^2) = aB/A = V_\infty$, using $AB = \delta/\lambda$. Both reductions were verified numerically against direct quadrature, as was (1.67) for general r , to relative error $\sim 10^{-8}$ in the convergent regime. Where the transforms are needed as functions of time rather than of r , the standard numerical inversions apply [21, 22].

The companion derivations — the lifetime yield $V_\infty^{(k)}$ and the generating-function coefficient extraction used by the chained model — are in the chapter on its distribution theory's technical appendix.

1.E Release-model catalogue

Section 1.6.1 names five ODE cases; sections 1.6.3 and 1.6.4 state what is settled about self-exciting and partial release. This appendix writes the cases out and records the candidates behind those statements.

1.E.1 Five ODE cases

Target cells T are fixed; $\mathcal{I}$ denotes productive cells, $\mathcal{V}$ free particles, E eclipse cells, Q the total intracellular stock. Production into the stock is mean-linear at rate ν per cell.

Case 1 — classical BMVR.

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (1.68)$$

Case 2 — eclipse and classical release.

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\omega + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \omega E - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (1.69)$$

$$R_0 = \frac{\gamma T}{c} \cdot \frac{\omega}{\omega + d_E} \cdot \frac{p}{d_{\mathcal{I}}}. \quad (1.70)$$

Case 3 — stock and continuous export.

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \varepsilon Q - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \varepsilon Q - c \mathcal{V}. \quad (1.71)$$

In the quasi-steady stock regime,

$$Q_* = \frac{\nu}{\varepsilon + d_{\mathcal{I}}} \mathcal{I}, \quad p_{\text{eff}} := \frac{\varepsilon Q_*}{\mathcal{I}} = \frac{\varepsilon \nu}{\varepsilon + d_{\mathcal{I}}}, \quad (1.72)$$

so classical BMVR is the fast-export limit of stock-and-export dynamics. Without eclipse,

$$R_0 = \frac{\gamma T \nu \varepsilon}{c d_{\mathcal{I}} (\varepsilon + d_{\mathcal{I}})}. \quad (1.73)$$

Case 4 — stock and proportional full clear.

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}, \quad (1.74)$$

The mature balance $\nu = \delta \bar{x}^2 + d_{\mathcal{I}} \bar{x}$ gives

$$\bar{x} = \frac{-d_{\mathcal{I}} + \sqrt{d_{\mathcal{I}}^2 + 4\delta\nu}}{2\delta}, \quad p_{\text{eff}} = \delta \bar{x}^2. \quad (1.75)$$

Case 5 — eclipse, stock and proportional full clear.

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\omega + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \omega E - d_{\mathcal{I}} \mathcal{I}, \quad (1.76)$$

$$\frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}, \quad R_0 = \frac{\gamma T}{c} \cdot \frac{\omega}{\omega + d_E} \cdot V_{\infty}. \quad (1.77)$$

1.E.2 Hataye limiting-dilution numbers

1.E.3 Self-exciting release

Twelve candidates, four developed below (table 1.14). They nest: Model 2, with intensity $\lambda(t) = f(X_t) + \sum_{t_i < t} h(t - t_i)$, is the general event-history form, and $h \equiv 0$ with $f(x) = \varepsilon x$ recovers continuous export (Case 3), $h \equiv 0$ with $f(x) = \delta x^2$ and full clear recovers Model 7 and Cases 4–5, an exponential $h(t) = \eta e^{-\zeta t}$ is equivalent to Model 10, and Model 6 is a coarse two-level version without a continuous intensity.

Quantity	Value
First detection, bulk production	day 3; days 3–5
Favoured eclipse stages n	≈ 5
CD4 division / eclipse death	$\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$
Detectable fraction f_{HIV}	0.41
Allee threshold Λ	2.3 latently infected cells
RNA copies at that threshold	5100
Establishment definition	$\geq 2 \times 10^5$ copies
Establishment from one cell	2%
RNA copies at inflection	$\sim 5 \times 10^3$

Table 1.13: Numbers quoted from Hataye et al. [17] for the HIV contrast of [section 1.6.5](#).

#	Model	Excitation carried by	ODE BMVR?	Renewal?
1	Classical Hawkes	event history only	poor	yes (numerical g)
2	Hawkes + stock X	events + depletion/refill	via Model 10	yes
3	Marked Hawkes	size-weighted events	approximate	yes (numerical)
4	ETAS-style	size + time kernel	approximate	yes (numerical)
5	Cox + latent Z	latent excitability	if Z simple	yes
6	ON/OFF telegraph	sustained high phase	yes	yes
7	$\lambda \propto X^2$	current load only	yes (Cases 4–5)	yes
8	Nucleation biophysics	assembly cascade	hard	yes (numerical)
9	Age since last release	renewal excitation	yes (stages)	yes
10	Jumping rate $\psi(t)$	shot-noise intensity	yes (mean-field)	yes
11	Spatial Hawkes	local membrane	average first	after coarse-graining
12	Full clear (Cases 4–5)	instant total cascade	yes	yes

Table 1.14: Twelve self-exciting release mechanisms. The last two columns are the practical question: every row admits a renewal description, because $g(\alpha) = \mathbb{E}[\lambda(\alpha) \mid \text{alive}]$ always exists; only some admit a finite ODE one [16].

Model 7 — cooperative intensity, $\lambda(x) = \delta x^2$ (or $\delta x(x - 1)$). No memory of past events, only the current load: production ν , optional degradation μX , full-clear dumps at rate δX^2 . The ODE form is Cases 4–5 under the homogeneous closure, the renewal form $g(\alpha) = \delta \mathbb{E}[X_\alpha^2 \mathbf{1}\{\text{alive}\}]$.

Model 6 — ON/OFF telegraph. The cell, or a membrane site, switches between OFF and ON at rates $\sigma_{\text{on}}, \sigma_{\text{off}}$, releasing at rate ρ , or εX , while ON: intermittent surges without full clear. The simplest ODE version splits the infected class:

$$\frac{dI_{\text{OFF}}}{dt} = \gamma T \mathcal{V} + \sigma_{\text{off}} I_{\text{ON}} - (\sigma_{\text{on}} + d_{\mathcal{I}}) I_{\text{OFF}}, \quad (1.78)$$

$$\frac{dI_{\text{ON}}}{dt} = \sigma_{\text{on}} I_{\text{OFF}} - (\sigma_{\text{off}} + d_{\mathcal{I}}) I_{\text{ON}}, \quad \frac{d\mathcal{V}}{dt} = \rho I_{\text{ON}} - c \mathcal{V}, \quad (1.79)$$

and quasi-steady switching gives effective budding at the duty cycle, $p_{\text{eff}} = \rho \sigma_{\text{on}} / (\sigma_{\text{on}} + \sigma_{\text{off}})$: a transparent limit back to constant p .

Model 10 — jumping continuous release rate. A continuous intensity $\psi(t) \geq 0$ jumps by η at each release event and decays to baseline ψ_0 at rate ζ , $d\psi = -\zeta(\psi - \psi_0)dt + \eta dN_t$, with flux $\psi(t)$ or $\psi(t)X_t$: the Markovian, shot-noise form of a Hawkes process with exponential kernel. Tracking $\mathcal{I}$, Q and $\Psi = \sum_i \psi_i$ with $\bar{\psi} = \Psi/\mathcal{I}$, a mean-field closure gives ODEs for $(\mathcal{I}, Q, \Psi, \mathcal{V})$

with flux $\approx \bar{\psi}Q$ — fading memory with a finite-dimensional state, at the cost of closures and three extra parameters.

Model 2 — Hawkes with stock. The literal version: $\lambda(t) = f(X_t) + \sum_{t_i < t} h(t - t_i)$; on an event X drops by the released amount, between events it refills at rate ν , the cell dies at rate $d_{\mathcal{I}}$. The renewal form $g(\alpha) = \mathbb{E}[\lambda(\alpha) \mid \text{alive}]$ is always well defined and typically numerical; an ODE form exists only after reduction to exponential h , that is to Model 10 (table 1.15).

	7 $\lambda \propto X^2$	6 ON/OFF	10 jumping ψ	2 Hawkes+ X
Carrier	load only	discrete phase	continuous ψ	event history (+load)
Memory	Markov in X	Markov in phase	Markov in ψ	non-Markov unless exp. h
Clusters	high $X \rightarrow$ fast dumps	long ON sojourn	high- ψ window	aftershocks
ODE BMVR	yes (Cases 4–5)	yes (split $\mathcal{I}$)	yes (mean-field Ψ)	only via 10
Renewal	yes	yes	yes	yes
$\rightarrow$ classic p	QSS $\delta\bar{x}^2$	duty cycle	QSS $\mathbb{E}[\psi X]$	as 10 if exp. h

Table 1.15: The four developed self-excitation models. Read the last row as the answer to when each collapses back to a constant release rate.

1.E.4 Partial release

At load n , a release event sends out B with $0 \leq B \leq n$ and retains $n - B$. The event clock $\lambda(X)$ is free — constant δ , linear δX , or cooperative δX^2 — and so is the law of $B \mid X = n$, for which there are seven options:

- A. **Boolean thinning**, the default: each unit released independently with probability φ , so $B \mid X = n \sim \text{Binomial}(n, \varphi)$; $\varphi \rightarrow 0$ gives tiny fractional clear, $\varphi = 1$ full clear, and generating functions stay tractable.
- B. Fixed fraction $B = \text{round}(fn)$: A's mean with $\varphi = f$, less stochasticity. **C.** Random fraction ($U \sim \text{Beta}$, or uniform thinning): more variable bursts than B.
- D. Fixed capped batch $B = \min(k, n)$: $k = 1$ is unit budding in the frequent-event limit, large k approaches full clear. **E.** Size-biased release, $\mathbb{P}\{B = j \mid n\} \propto j$: natural if one site releases a random assembled blob.
- F. Threshold-then-partial: events only for $X \geq x_*$, then A or B. **G.** Short high-export window: a triggered burst of high ε , partial clear in expectation, related to Model 6.

The single-cell boolean process produces at rate ν , fires release events at rate $\lambda(X)$ with $B \sim \text{Binomial}(X, \varphi)$ released and $X \rightarrow X - B$, and dies at rate $d_{\mathcal{I}}$; its nested limits are table 1.16. Option A with a load-dependent clock is the intermediate used in section 1.6.3: one transparent parameter between mild partial clear and full dumping, mechanistic, nested in Cases 3–4 at the mean-field level, and straightforward to simulate.

Choice	Recovers
$\varphi = 1, \lambda = \delta X^2$	full clear, cooperative (Case 4 style)
$\varphi = 1, \lambda = \delta X$	full clear, linear hazard
$0 < \varphi < 1, \lambda = \delta X$	the main partial-clear intermediate
$B \equiv 1, \lambda = \varepsilon X$	unit budding (Case 3, discrete)

Table 1.16: What the boolean process recovers at its corners. The middle row is the only one that is not already somewhere else in the chapter.

1.F The HIV stage model

Section 1.6.5 argues that HIV is not a birth–death–catastrophe process but that the skeleton (1.13) still carries it, on kernels $(S_{\text{HIV}}, g_{\text{HIV}})$ supplied by an eclipse-structured short-productive-pulse model. This appendix records that model explicitly. It is a candidate, not a fitted result: the point is to fix which system is meant by figure 1.15.

Write the system as follows. Let L_0 be a stimulated latently infected cell; two eclipse pathways run in parallel, A of high release potential and multi-stage, B low and brief:

$$L_0 \xrightarrow{\alpha_A} E_1^{(A)}, \quad L_0 \xrightarrow{\alpha_B} E^{(B)}, \quad E_j^{(A)} \xrightarrow{\omega} E_{j+1}^{(A)}, \quad E_n^{(A)} \xrightarrow{\omega} I^{(A)}, \quad E^{(B)} \xrightarrow{\alpha_B} I^{(B)}. \quad (1.80)$$

Eclipse cells divide or die without releasing virus,

$$E \xrightarrow{\rho_{\text{div}}} E + E, \quad E \xrightarrow{\mu_E} \emptyset, \quad (1.81)$$

with $\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$; productive cells release at constant rate and die,

$$I^{(A)} \xrightarrow{p_A} I^{(A)} + \mathcal{V}, \quad I^{(A)} \xrightarrow{\delta_I} \emptyset, \quad I^{(B)} \xrightarrow{p_B} I^{(B)} + \mathcal{V}, \quad I^{(B)} \xrightarrow{\delta_I} \emptyset, \quad (1.82)$$

with $p_A \gg p_B$ matching the bimodal release distribution and free virus clearing at rate c . Suppressing the dual path, the stage equations are

$$\frac{dE_j}{dt} = \omega E_{j-1} + \rho_{\text{div}} E_j - (\mu_E + \omega) E_j, \quad j = 1, \dots, n, \quad \frac{dI^{(A)}}{dt} = \omega E_n - \delta_I I^{(A)}, \quad (1.83)$$

with $E_0 \equiv 0$, the initial cohort entering as $E_1(0) = \alpha_A L_0$, and $\mathcal{V}' = p_A I^{(A)} - c\mathcal{V}$ ($+p_B I^{(B)}$). Division in eclipse is a net gain of $+\rho_{\text{div}} E_j$ and appears once, not on both sides.

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